

ON THE STRUCTURE OF COMPLEX SPECTRA AND EIGENFUNCTIONS OF TRANSFER AND KOOPMAN OPERATORS

MATHEUS M. CASTRO AND GARY FROYLAND

ABSTRACT. Complex spectra of transfer and Koopman operators describe rotational motion in dynamical systems. A particularly relevant situation in applications is where the rotation speed depends on the position in the phase space. We consider a canonical model of such dynamics in the presence of small noise, and provide precise characterisations of the eigenspectrum and eigenfunctions of the corresponding transfer operators. Further, we study the limiting behaviour of the eigenspectrum and eigenfunctions in the zero-noise limit, including their quadratic and linear response. Our results clarify the structure of transfer and Koopman operator eigenspectra, and provide new interpretations for applications. Our theorems on support localisation of the eigenfunctions yield simple algorithms to detect the existence and phase-space location of approximately cyclic motion with distinct periods. Our response results demonstrate that information on the cycle periods and their locations determined by the operator eigendata is largely insensitive to the noise level. We believe the mechanisms creating the eigendata apply more broadly and enhance our understanding of approximate cycle detection in dynamical systems with operator methods.

1. INTRODUCTION

Over the last three decades, operator-theoretic techniques have proven to be highly successful methods of analysis of complex dynamical systems. The basis of the analysis is usually the dominant eigenspectrum of a transfer operator or Koopman operator and the corresponding eigenfunctions. In the autonomous setting, large real spectral values correspond to almost-invariant sets [6, 5], a concept that has been applied extensively in several fields, including molecular dynamics [23], space mission design [4], and physical oceanography [12, 13, 21]. Regarding complex eigenvalues, the Halmos–von Neumann Theorem (see e.g. Theorem 17.11 [7]) states that each ergodic measure-preserving system with discrete spectrum is isomorphic to an ergodic rotation system. This concept has been used extensively in the context of Koopman operators, e.g. [20, 22].

The current paper takes the underlying Halmos–von Neumann idea and extends it in two ways. Firstly, we allow the rotation to depend on the location in phase space, rather than being fixed along a particular coordinate throughout phase space. Secondly, we make a detailed study of the effect of noise on the complex eigenvalues and eigenfunctions. The motivation for this study is that complex dynamical systems often possess many co-existing cycles or approximate cycles. For example, in the Earth system, climate drivers such as the El-Niño Southern Oscillation (ENSO) and the Madden–Julien Oscillation (JPO) are approximate co-existing cycles that influence the weather and climate. On even longer timescales, changing glacial cycles describe how northern hemisphere continental ice sheets repeatedly accumulate and melt over millions of years.

SCHOOL OF MATHEMATICS AND STATISTICS, UNSW SYDNEY, SYDNEY NSW 2052, AUSTRALIA

E-mail address: m.manzatto_de_castro@unsw.edu.au, g.froyland@unsw.edu.au.

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Recent work [9, 10] has demonstrated that transfer operator methods are a flexible and reliable way of extracting cycles from models, observations, and even time series of spatial fields; for example extracting the ENSO cycle from a time series of observed sea-surface temperature fields. Long-lived persistent cycles typically give rise to complex eigenvalues in the dominant spectrum of the transfer operator. The magnitude of the eigenvalue quantifies the rate of temporal decay of the cycle, while the argument of the eigenvalue directly provides an estimate of the cycle’s period. Moreover, the corresponding eigenfunctions may be used to represent the cycle in terms of the original spatiotemporal time series [9].

These ideas may be illustrated with the Lorenz equations, where an estimate of the transfer operator is constructed and the complex eigenvalue with largest magnitude is computed. The left panel of Figure 1 displays the magnitude of the complex entries of the corresponding eigenvector on a blue colourscale, and the right panel shows the argument of the complex entries on a periodic colourscale. In this example, the argument of the

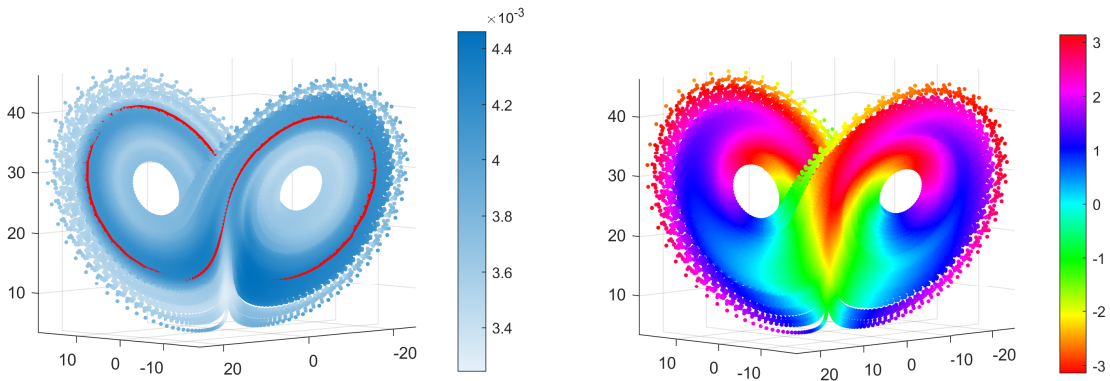


Figure 1. Left: Magnitude of the leading complex eigenvector of the normalised transfer operator for the Lorenz flow. A peak in magnitude occurs in the vicinity of the lowest-period periodic orbit of the Lorenz equations with a period of 1.5586 time units (shown in red, see Fig. 10a [11] and [8]). Right: The argument of the complex eigenvector. In the periodic colormap, one sees a full rotation around each of the two wings as the argument proceeds from $-\pi$ to π ; see [9].

eigenvalue associated with this complex eigenvector corresponds to a cycle of period 0.770. This period is roughly half of the period of the lowest-period periodic orbit of the Lorenz equations (displayed in red in Figure 1(left)). We make this comparison to emphasise that we are *not* aiming to detect single periodic orbits, but instead to capture large regions in phase space that revolve with an approximately constant period. The eigenvector detects that the dominant cycle in the Lorenz system consists of wide bands on each wing that rotate around each wing with a period of roughly 0.770. This duration happens to be approximately half of the period of the shortest periodic orbit, which loops once around each wing. The symmetry of the Lorenz equations under $(x, y, z) \mapsto (-x, -y, z)$, is reflected in the representation of the cycle in Figure 1(right); see [9] for details.

Our focus in this work is to provide a formal explanation for the behaviour of the spectrum and eigenfunctions of the transfer operator in the presence of coexisting cycles with different periods. Roughly speaking, eigenfunctions corresponding to cycles with distinct periods should have supports that are largely disjoint, because large overlaps would need to simultaneously rotate at two distinct speeds. Our formal analysis concentrates on a mathematical model with multiple coexisting cycles, acting on a discrete collection of

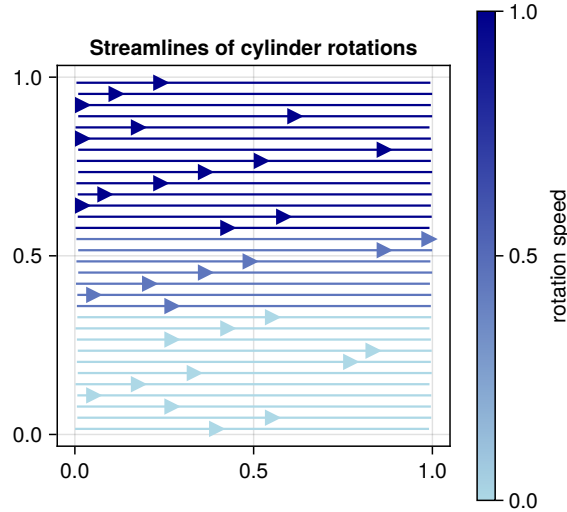


Figure 2. Streamlines of the cylinder rotation model with three bands of circular fibres with widths $L_1 = 11, L_2 = 7, L_3 = 15$ and rotation speeds $\beta_1 = \pi/20, \beta_2 = e/7, \beta_3 = 1/\sqrt{2}$, respectively.

circles, between which trajectories may randomly jump under the influence of noise. Our phase space M is the union of N circles: $M = \{1, \dots, N\} \times \mathbb{S}^1$, which we view as a cylinder that has been discretised along its central axis. The dynamics on the j^{th} circular band of the discretised cylinder is a rotation with speed α_j . We are particularly interested in the case where several neighbouring circular fibres rotate with the same speed; see Figure 2. The dynamics on these groups of fibres model bands of fluid rotating with a common speed.

The addition of noise is an important selection procedure that promotes significant eigenfunctions of the transfer/Koopman operator by associating them with large-magnitude eigenvalues. This heuristic is very classical for the eigenvalue 1 where the eigenfunction corresponds to the invariant measure [18, 17, 24, 1], and classical for large positive real eigenvalues in connection with almost-invariant sets and metastability [6]. Moreover, the addition of noise may also be useful for proving convergence of numerical schemes [6, 16, 3]. It has also been applied in the context of thermodynamic formalism, where noise perturbations identify equilibrium states by associating them with large-magnitude eigenvalues of suitable operators [2]. Further motivations for considering the addition of noise are:

- (1) models and observations are imperfect and often subject to small errors;
- (2) most numerical approaches to estimating transfer and Koopman operators introduce artificial noise or diffusion through reduction to a finite-rank representation of the operator;
- (3) the addition of noise enables formal discussion of spectra and eigenfunctions of transfer operators acting on simple spaces such as L^2 ;
- (4) zero-noise limits are difficult limits of singular type, yet remain relevant and are the type observed in numerical experiments.

This work builds fundamental theory to support the heuristic notion that the addition of noise *promotes eigenfunctions connected with dominant cycles toward the largest-magnitude complex eigenvalues*. We achieve this by conducting a formal analysis of a

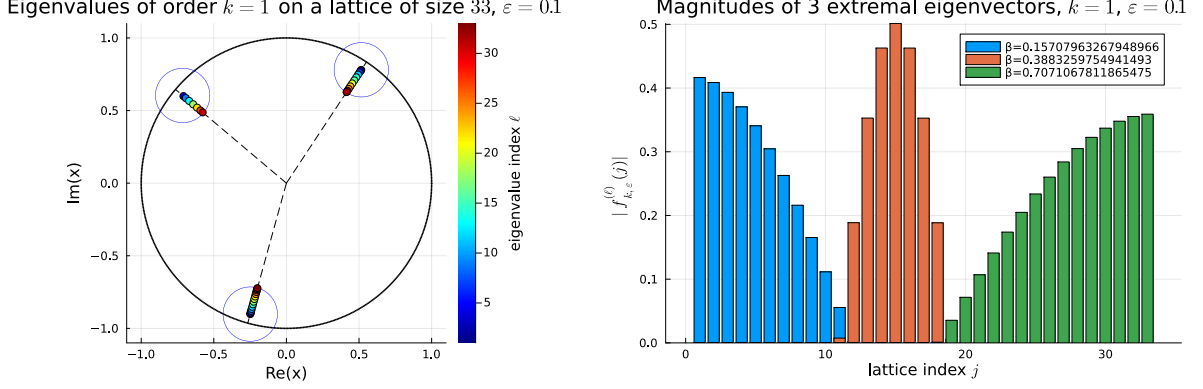


Figure 3. *Left:* The small coloured disks indicate the spectrum $\lambda_{1,\varepsilon}$ (for $\delta = \varepsilon = 0.1$) of the 33×33 matrix $D_{k,\alpha}W_\varepsilon = D_{k,\beta,L}W_\varepsilon$, associated to the model from Figure 1, where $\beta = (\pi/20, e/7, 1/\sqrt{2})$ and $L = (11, 7, 15)$. The large black circle is the unit circle in the complex plane. Note that the 33 eigenvalues are distinct and that they appear in groups, with L_i eigenvalues nearby $\exp(-2\pi i k \beta_i)$ for $i = 1, 2, 3$. The relevant bounds (6) from Proposition 2.2 are shown as dark blue circles. *Right:* Plot of $|f_{k,\varepsilon}^{(\ell)}(j)|$ vs lattice index j for $\ell = 1, 2, 4$, where the ordering of the eigenvalues are shown in the left panel. We plot the eigenvectors of the largest-magnitude eigenvalue from each of the three groups; the eigenvectors have been normalised to have unit norm.

model system that possesses the relevant cyclic properties. We perturb this model with noise and carefully study the zero noise limit, fully characterising the limiting eigenvalues and the form of the limiting eigenfunctions. Moreover, we prove quadratic response of the eigenvalues and linear response of the eigenfunctions, the form of which demonstrates a strong robustness of the eigenfunctions to a range of noise.

To introduce noise to our model we use a symmetric random walk between the circular fibres with weights given by a symmetric $N \times N$ doubly-stochastic matrix $W_\varepsilon \geq 0$ that converges to the identity matrix as $\varepsilon \rightarrow 0$. On each circular fibre we apply i.i.d. additive noise. In summary, the dynamics $T : M \rightarrow M$ is given by $T(j, x) = (j + \gamma, x + \alpha_j + \eta)$, where γ is an integer-valued random variable that generates a random walk on $\{1, \dots, N\}$ with reflecting boundary conditions at $j = 1$ and $j = N$, and η is an i.i.d. random variable with values uniformly distributed in $[-\delta, \delta]$.

For this model, we demonstrate a separable structure of the eigenfunctions (Proposition 2.2) of the transfer operator and use this structure to generate a series of precise results regarding zero-noise limits. We prove that as the noise ε approaches zero, the eigenvalues converge to a complex exponential given by one of the rotation speeds, at an explicit rate linear in ε (Proposition 2.2). When there are bands of neighbouring fibres that share a rotation speed, clusters of eigenvalues approach a single complex exponential (Theorem 2.5 and Figure 3(left)). We prove that in the $\varepsilon \rightarrow 0$ limit, the corresponding eigenfunctions approach an orthonormal basis supported on the band (Theorem 2.5 and Figure 3(right)). Despite having multiple eigenvalues converging to a single eigenvalue, we develop a quadratic response formula for the eigenvalues and linear response formulae for the eigenfunctions (Theorem 2.6). The supports of the latter is also restricted to the band, resulting in this support localisation property being particularly robust for small noise levels. Proposition 2.7 covers the response of perturbations with respect to the rotation speeds $\{\alpha_j\}_{j=1}^N$.

An outline of the paper is as follows. We construct our model and state our main results in Section 2. The proofs of the main results are carried out in Sections 3–6. Section 7 provides a detailed case study illustrating and interpreting our main results for the example presented in Figure 2. Figure 3 summarises the behaviour of the most important (lowest-order) eigenvalues and eigenvectors for modest positive ε , Figure 5 summarises the response behaviour of these eigenvalues and eigenvectors, and Figure 4 visualises the magnitude and argument of the corresponding eigenfunctions on the full cylindrical domain.

2. MODEL AND STATEMENTS OF MAIN RESULTS

2.1. The model. The Perron–Frobenius operator for the random walk is given by

$$\mathcal{P}_{\text{base},\varepsilon}f(j) := \sum_{\gamma \in \pm\{0,1,\dots,N\}} W_{\varepsilon,j,j+\gamma} f(j-\gamma), \quad (1)$$

with the convention that if $j+\gamma \notin \{1,\dots,N\}$, then $W_{\varepsilon,j,j+\gamma} = 0$. On the j^{th} band, the annealed transfer operator $\mathcal{P}_{j,\delta} : L^1(\mathbb{S}^1) \rightarrow L^1(\mathbb{S}^1)$ is given by $\mathcal{P}_{j,\delta}h(x) := \frac{1}{2\delta} \int_{[-\delta,\delta]} h(x - \alpha_j - \eta) d\eta$.

The product of the discrete uniform probability measure on $\{1,\dots,M\}$ and normalised Lebesgue measure on $\mathbb{S}^1$ is denoted by m . Using the form of T , and letting $F \in L^1(M)$, the annealed transfer operator with respect m is given by

$$\mathcal{P}_{\varepsilon,\delta}F(j,x) = \sum_{\gamma \in \pm\{0,1,\dots,N\}} W_{\varepsilon,j,j+\gamma} \frac{1}{2\delta} \int_{[-\delta,\delta]} F(j-\gamma, x - \alpha_{j-\gamma} - \eta) d\eta.$$

We now write $F(j,x) = f(j)h_j(x)$ to highlight the skew-product nature of T and $\mathcal{P}_{\varepsilon,\delta}$.

$$\begin{aligned} \mathcal{P}_{\varepsilon,\delta}F(j,x) &= \sum_{\gamma \in \pm\{0,1,\dots,N\}} W_{\varepsilon,j,j+\gamma} f(j-\gamma) \frac{1}{2\delta} \int_{[-\delta,\delta]} h_{j-\gamma}(x - \alpha_{j-\gamma} - \eta) d\eta \\ &= \sum_{\gamma \in \pm\{0,1,\dots,N\}} W_{\varepsilon,j,j+\gamma} f(j-\gamma) \mathcal{P}_{j-\gamma,\delta}h_{j-\gamma}(x) \\ &= \mathcal{P}_{\text{base},\varepsilon}(f(j)\mathcal{P}_{j,\delta}h_j(x)). \end{aligned} \quad (2)$$

Because each $\mathcal{P}_{j,\delta}$ is the transfer operator for a rotation on the j^{th} circle by α_j (plus uniform bounded noise), the eigenfunctions of $\mathcal{P}_{j,\delta}$ are complex exponentials of increasing order, *independent of j* . Thus, as we shall shortly see, the eigenfunctions of $\mathcal{P}_{\varepsilon,\delta}$ are *separable in the variables j and x* .

2.2. General behaviour of eigenvalues and eigenfunctions.

2.2.1. The case of $\varepsilon = 0$. Before we begin our main analysis, it is informative to consider the simple situation where $\varepsilon = 0$ and $\delta \geq 0$. With no noise connecting the distinct copies of $\mathbb{S}^1$, the rotations on each circle all individually contribute to generate the full eigenspectrum of $\mathcal{P}_{0,\delta}$ consisting of the N -fold union of the eigenvalues $\bigcup_{j=1}^N \sigma(\mathcal{P}_{j,\delta})$, where

$$\sigma(\mathcal{P}_{j,\delta}) = \bigcup_{k \in \mathbb{Z}} \left\{ \frac{\sin(2\pi k\delta)}{2\pi k\delta} e^{-2\pi i k \alpha_j} \right\} \cup \{0\}. \quad (3)$$

Indeed, in this case $\mathcal{P}_{\text{base},0} = \text{Id}$, therefore (2) implies that given $F(j, x) = \sum_{k \in \mathbb{Z}} f_k(j) e^{2\pi i k x} \in L^2(M)$, it implies that

$$\mathcal{P}_{0,\delta} F(j, x) = \sum_{k \in \mathbb{Z}} f(j) \mathcal{P}_{j,\delta}(e^{2\pi i k x}) = \sum_{k \in \mathbb{Z}} \left(\frac{\sin(2\pi k \delta)}{2\pi k \delta} e^{-2\pi i k \alpha_j} \right) f(j) e^{2\pi i k x},$$

where $\sin(0)/0$ should be interpreted as 1 in the above equation. Therefore, by comparing the Fourier modes of $\mathcal{P}_{0,\delta} F$ and λF we obtain (3).

Remark 2.1. All of the results in Section 2 can be easily generalized from the state space $M = \{1, \dots, N\} \times S^1$ to a state space $M = \{1, \dots, N\} \times \mathbb{T}^d$ where:

- (1) the α_j -rotations on the circle $\mathbb{S}^1 = \mathbb{T}^1$ are replaced with $(\alpha_{j,1}, \dots, \alpha_{j,d})$ -translations on the d -torus $\mathbb{T}^d$.
- (2) the discrete fibres $\{1, \dots, N\}$ are no longer discretely filling an interval, but discretely fill a connected subset of $\mathbb{R}^{d'}$.

Such a situation allows one to create models of rather general rotational dynamics. For example, a simple extension to $d = 1$ and $d' = 2$ would model fluid rotation around a solid three-dimensional torus, with the N fibres discretely filling out the non-periodic two-dimensional cross-section of the solid torus. Subsets of $\{1, \dots, N\}$ called *bands* (see Definition 2.3) represent solid “tubes” of three-dimensional fluid that rotate at a common angular speed inside a solid three-dimensional torus. Alternatively – remaining in a solid three-dimensional torus – we may set $d = 2$ and $d' = 1$ to model translations on two-dimensional nested toral shells within the solid torus. Bands (subsets of $\{1, \dots, N\}$) represent nested families of solid tori (like Matryoshka dolls) with common translations on each member of the family.

In this general setting, the spectrum of $\mathcal{P}_{0,\delta}$ is

$$\bigcup_{j=1}^N \sigma(\mathcal{P}_{j,\delta}) = \bigcup_{j=1}^N \bigcup_{k \in \mathbb{Z}^d} \left\{ \prod_{q=1}^d \frac{\sin(2\pi k_q \delta)}{2\pi k_q \delta} e^{-2\pi i k_q \alpha_{j,q}} \right\} \cup \{0\}.$$

Concerning the extension from S^1 to $\mathbb{T}^d$, the same proofs of all of our results follow with minor adaptations. For the extension of the fibres $\{1, \dots, N\}$ filling out sets in $\mathbb{R}^{d'}$, one only needs to appropriately adjust the matrices W_ϵ . For example, in d' dimensions, a natural choice would be to construct W_ϵ from a finite-difference stencil for the Laplace operator on a regular grid in d' dimensions (see (43)–(44) for such a choice for $d' = 1$). For the sake of simplicity of exposition, we restrict the statements and proofs of all results to the case $d = 1$.

2.2.2. The case of $\epsilon > 0$. In the remainder of this work, we will assume that $\delta = 0$ because

- (1) the effect of δ on the spectrum is simply to multiply each eigenvalue by $\frac{\sin(2\pi k \delta)}{2\pi k \delta}$, and
- (2) the eigenfunctions are unaffected by changes in δ .

Hereafter we will denote $\mathcal{P}_{\epsilon,\delta}$ by simply $\mathcal{P}_\epsilon$. When $\epsilon > 0$ we no longer have a closed form expression for the eigenvalues and eigenfunctions of $\mathcal{P}_\epsilon$, but we can characterise them as eigenvalues and eigenvectors of the simple matrix product $D_{k,\alpha} W_\epsilon$ where

$$D_{k,\alpha} = \text{Diag}(e^{-2\pi i k \alpha_1}, \dots, e^{-2\pi i k \alpha_N}), \quad (4)$$

and analyse their behaviour for small $\epsilon > 0$ as $\epsilon \rightarrow 0$.

Proposition 2.2. *Let $\alpha = (\alpha_1, \dots, \alpha_N) \in \mathbb{R}^N$. For each fixed $k \in \mathbb{Z}$:*

- (1) For every $\varepsilon > 0$, the operator $\mathcal{P}_\varepsilon : L^2(M) \rightarrow L^2(M)$ has N eigenvalues $\lambda_{k,\varepsilon}^{(\ell)}$, $\ell = 1, \dots, N$, and N (generalised) eigenfunctions $F_{k,\varepsilon}^{(\ell)}$, of the separable form

$$F_{k,\varepsilon}^{(\ell)}(j, x) = f_{k,\varepsilon}^{(\ell)}(j) e^{2\pi i k x} \quad \text{for all } \ell \in \{1, \dots, N\}. \quad (5)$$

- (2) For sufficiently small $\varepsilon > 0$ one has

$$|\lambda_{k,\varepsilon}^{(\ell)} - e^{-2\pi i k \alpha_\ell}| \leq 2 \max_{1 \leq \ell \leq N} |\dot{W}_{\ell\ell}| \varepsilon \quad \text{for all } \ell \in \{1, \dots, N\}, \quad (6)$$

Proof. See Section 3.1. $\square$

Using Proposition 2.2, we assume from now on that the eigenvalues $\lambda_{k,\varepsilon}^{(1)}, \dots, \lambda_{k,\varepsilon}^{(N)}$ are labelled such that

$$\lambda_{k,\varepsilon}^{(\ell)} \xrightarrow{\varepsilon \rightarrow 0} e^{-2\pi i k \alpha_\ell} \quad \text{for every } \ell \in \{1, \dots, N\}. \quad (7)$$

2.3. The case of multiple fibres with common rotation speed. A particularly interesting situation arises when some α_j are identical for neighbouring j s. This corresponds to a band of fluid rotating at the same speed, and handling this physically relevant situation requires a more careful analysis. This motivates the introduction of *bands* below.

Definition 2.3 (*S*-banded). A N -vector $\alpha = (\alpha_1, \dots, \alpha_N) \in \mathbb{R}^N$ is said to be *S*-banded if there exist

- (i) a natural number $1 \leq S \leq N$,
 - (ii) distinct real numbers $\beta_1, \dots, \beta_S \in \mathbb{R}$, and
 - (iii) a vector of positive integer numbers $L = (L_1, \dots, L_S)$ satisfying $\sum_{i=1}^S L_i = N$,
- such that

$$\alpha_j = \beta_s \quad \text{if } N_{s-1} < j \leq N_s \quad \text{for some } s \in \{1, \dots, S\},$$

where $N_0 < N_1 < \dots < N_S = N$ is a sequence of recursively defined natural numbers

$$N_0 = 0 \text{ and } N_i = L_i + N_{i-1} \text{ if } i \in \{1, \dots, S\}$$

In the above context, the vector $\beta = (\beta_1, \dots, \beta_S)$ is referred to the *band vector*, and L is called the *band-width vector*. Moreover, given $s \in \{1, \dots, S\}$ we denote the *band members of β_s* as $B_s = \{j \in \mathbb{N} : N_{s-1} < j \leq N_s\}$. Observe that from the convention adopted in (7) we have that $\{\ell \in \{1, \dots, N\} : \lambda_{k,\varepsilon}^{(\ell)} \rightarrow e^{-2\pi i k \beta_s} \text{ as } \varepsilon \rightarrow 0\} = B_s$.

Note that a vector $\alpha \in \mathbb{R}^N$ with distinct entries is N -banded. For $S < N$, there is at least one $1 \leq s \leq S$ such that β_s has band width $L_s \geq 2$. Proposition 2.2(2) tells us that for a given Fourier index k , $\lambda_{k,\varepsilon}^{(\ell)}$ converges to $\exp(-2\pi i k \beta_s)$ for L_s -fold ℓ , this phenomenon is illustrated in Figure 3 (left). Therefore, one expects an L_s -dimensional subspace spanned by the corresponding eigenvectors $F_{k,\varepsilon}^{(\ell)}$ to converge to some L_s -dimensional space, but the question of convergence of individual eigenvectors and their characterisation is unclear.

Definition 2.4 (*L*-admissible family of matrices). The family of $N \times N$ matrices $\{W_\varepsilon = \text{Id} + \varepsilon \dot{W}\}_{\varepsilon \geq 0}$ is *L*-admissible if the symmetric matrix $\dot{W}$ satisfies:

- (1) $\dot{W}_{ij} \geq 0$ for $i \neq j \in \{1, \dots, N\}$ and $\sum_{j=1}^N \dot{W}_{ij} = 0$, $\forall i \in \{1, \dots, N\}$.
- (2) $\dot{W}$ has N distinct eigenvalues.
- (3) For each $s \in \{1, \dots, S\}$, the $L_s \times L_s$ submatrix $\hat{W}_s := [\dot{W}_{jk}]_{N_{s-1} < j, k \leq N_s}$ has L_s distinct eigenvalues.

In the $\varepsilon \rightarrow 0$ limit, instead of considering eigenfunctions, we use eigenprojections, as eigenfunctions can always be multiplied by unit-norm complex scalars, which alters their phase. This phase of freedom can destroy both convergence and linear response. Eigenprojections, on the other hand, remain well-defined and invariant under such transformations. Theorem 2.5 establishes that if the elements of the band vector are incommensurate, the eigenprojections of $\mathcal{P}_\varepsilon$ converge as $\varepsilon \rightarrow 0$. Furthermore, the limiting projections can be precisely characterized in terms of eigenvectors of specific matrices.

Theorem 2.5. *Let $N \in \mathbb{N}$, $1 \leq S \leq N$, $L = (L_1, \dots, L_S)$ be a band-width vector and $\{W_\varepsilon = \text{Id} + \varepsilon \dot{W}\}_{\varepsilon \geq 0}$ an L -admissible family of matrices. Then for each $\beta \in \Gamma := \{(\beta_1, \dots, \beta_S) \in \mathbb{R}^S : e^{-2\pi i k \beta_{s_1}} \neq e^{-2\pi i k \beta_{s_2}} \ \forall k \in \mathbb{Z}, \forall s_1, s_2 \in \{1, \dots, S\}\}$ and $k \in \mathbb{Z}$:*

- (1) *For sufficiently small ε (depending on k), the N generalised eigenfunctions $F_{k,\varepsilon}^{(\ell)}$ from (5) become N eigenfunctions with distinct eigenvalues $\lambda_{k,\varepsilon}^{(\ell)}$, $\ell = 1, \dots, N$.*
- (2) *For each $\varepsilon > 0$ sufficiently small and each $\ell = 1, \dots, N$, recall $F_k^{(\ell)}(j, x) = f_k^{(\ell)}(j) e^{2\pi i k j x}$ and let $\Pi_{k,\varepsilon}^{(\ell)}, \Pi_k^{(\ell)} : L^2(M) \rightarrow L^2(M)$ denote L^2 -orthogonal projections onto the spans of the eigenfunctions $F_{k,\varepsilon}^{(\ell)}$ and $F_k^{(\ell)}$, respectively. In the $\varepsilon \rightarrow 0$ limit there exists a limiting orthonormal basis $\{f_k^{(\ell)}\}_{\ell=1}^N$ such that*

$$\Pi_{k,\varepsilon}^{(\ell)} \xrightarrow{\varepsilon \rightarrow 0} \Pi_k^{(\ell)} \text{ in the } L^2(M)\text{-operator norm.}$$

We may further characterise the $\{f_k^{(\ell)}\}_{\ell=1}^N$ as follows:

- (a) *If $S = 1$ or $k = 0$, $\{f_k^{(\ell)}\}_{\ell=1}^N$ is an orthonormal eigenbasis of the $N \times N$ matrix $\dot{W}$.*
- (b) *If $S > 1$ and $k \in \mathbb{Z} \setminus \{0\}$, $\{f_k^{(\ell)}\}_{\ell=1}^N$ is an orthonormal eigenbasis of the matrix*

$$\hat{P}_{k,\beta,L} := D_{k,\beta,L} \hat{W}_L \tag{8}$$

where

$$D_{k,\beta,L} := \begin{pmatrix} e^{-2\pi i k \beta_1} \text{Id}_{L_1} & & \\ & \ddots & \\ & & e^{-2\pi i k \beta_S} \text{Id}_{L_S} \end{pmatrix}, \tag{9}$$

$$\hat{W}_L := \begin{pmatrix} \hat{W}_1 & & & \\ & \hat{W}_2 & & \\ & & \ddots & \\ & & & \hat{W}_{S-1} & \\ & & & & \hat{W}_S \end{pmatrix},$$

and Id_n is the $n \times n$ identity matrix. Moreover, for each $s \in \{1, \dots, S\}$, if ℓ is a band member of β_s , the support of $f_k^{(\ell)}$ is contained in $\{N_{s-1} + 1, \dots, N_s\}$.

Proof. See Sections 3.2, 3.3 and 4. □

2.4. Quadratic and linear response. Proposition 2.2 tells us that for each Fourier index $k \in \mathbb{Z}$, in the $\varepsilon \rightarrow 0$ limit the eigenvalues $\lambda_{k,\varepsilon}^{(\ell)}$, $\ell = 1, \dots, N$ of $\mathcal{P}_\varepsilon$ each converge to $e^{-2\pi i k \alpha_\ell}$, while Theorem 2.5 says that the eigenprojections $\Pi_{k,\varepsilon}^{(\ell)}$ associated with the eigenfunctions $F_{k,\varepsilon}^{(\ell)}$ of $\mathcal{P}_\varepsilon$ converge to L^2 -orthogonal projections $\Pi_k^{(\ell)}$ onto $\text{span}\{F_k^{(\ell)}\}$ which have explicit eigenfunction characterisations. A fundamental question arising from these results is the rate of convergence of $\Pi_{k,\varepsilon}^{(\ell)}$ and $\lambda_{k,\varepsilon}^{(\ell)}$ as $\varepsilon \rightarrow 0$. If all α_ℓ are distinct,

one may hope that the corresponding one-dimensional eigenprojections $\Pi_{k,\varepsilon}^{(\ell)}$ are regular functions of ε at $\varepsilon = 0$. On the other hand, if we have nontrivial bands – i.e. $S < N$ in Definition 2.3 – then we would expect at best regular dependence of the various L_s -dimensional eigenspaces, corresponding to the L_s repeated eigenvalues $e^{-2\pi i k \beta_s}$, at $\varepsilon = 0$. Surprisingly, in the nontrivially banded situation, we can in fact demonstrate quadratic response of *all individual eigenvalues* $\varepsilon \mapsto \lambda_{k,\varepsilon}^{(\ell)}$ at $\varepsilon = 0$, and linear response of *all individual eigenprojections* $\varepsilon \mapsto \Pi_{k,\varepsilon}^{(\ell)}$ at $\varepsilon = 0$. To further illustrate why this is surprising, we recall that the standard approach to establishing linear response of the eigenprojections is to differentiate the equation $\mathcal{P}_\varepsilon \Pi_{k,\varepsilon}^{(\ell)} = \lambda_{k,\varepsilon}^{(\ell)} \Pi_{k,\varepsilon}^{(\ell)}$ with respect to ε , where we use the notation $\dot{\cdot}$ to denote $\frac{\partial}{\partial \varepsilon}$. This differentiation yields the identity

$$\mathcal{P}_\varepsilon \dot{\Pi}_{k,\varepsilon}^{(\ell)} + \dot{\mathcal{P}}_\varepsilon \Pi_{k,\varepsilon}^{(\ell)} = \dot{\lambda}_{k,\varepsilon}^{(\ell)} \Pi_{k,\varepsilon}^{(\ell)} + \lambda_{k,\varepsilon}^{(\ell)} \dot{\Pi}_{k,\varepsilon}^{(\ell)},$$

which can be rewritten as

$$\dot{\Pi}_{k,\varepsilon}^{(\ell)} = - \left(\mathcal{P}_\varepsilon - \lambda_{k,\varepsilon}^{(\ell)} \right)^{-1} \left(\dot{\mathcal{P}}_\varepsilon - \dot{\lambda}_{k,\varepsilon}^{(\ell)} \right) \Pi_{k,\varepsilon}^{(\ell)}. \quad (10)$$

A common strategy is to evaluate equation (10) at $\varepsilon = 0$. However, in our setting, this approach fails because the range of the operator

$$\lim_{\varepsilon \rightarrow 0} \left(\dot{\mathcal{P}}_\varepsilon - \dot{\lambda}_{k,\varepsilon}^{(\ell)} \right) \Pi_{k,\varepsilon}^{(\ell)} \Big|_{\varepsilon=0}$$

lies outside the domain of $(\mathcal{P}_\varepsilon - \lambda_{k,\varepsilon}^{(\ell)})^{-1}$. To overcome this difficulty, we use a non-trivial cancellation mechanism specific to our framework. This allows us to obtain explicit expressions for the response terms for both the eigenvalues and eigenprojections.

Further, we observe numerically (see the case study in Section 7) that the form of the response of the eigenfunctions $\dot{F}_k^{(\ell)}$ is such that the overall shape of the envelope of $F_{k,\varepsilon}^{(\ell)}$ is relatively unchanged by modestly increasing ε from 0. Practically, this observation helps to explain the robustness of dominant complex eigenfunctions to various types of noise, whether they arise from models, data, or numerical techniques.

The following theorem refines Theorem 2.5 by presenting the linear response of the eigenprojections, in addition to the linear and quadratic responses of the eigenvalues of $\mathcal{P}_\varepsilon$ as $\varepsilon \rightarrow 0$.

Theorem 2.6. *Under the hypotheses and notation of Theorem 2.5, there exist complex numbers $\hat{\lambda}_k^{(1)}, \dots, \hat{\lambda}_k^{(N)}, \hat{\lambda}_k^{(1)}, \dots, \hat{\lambda}_k^{(N)}$ and functions*

$$f_k^{(1)}, \dots, f_k^{(N)}, \hat{f}_k^{(1)}, \dots, \hat{f}_k^{(N)} : \{1, \dots, N\} \rightarrow \mathbb{C},$$

such that

$$\lambda_{k,\varepsilon}^{(\ell)} = e^{-2\pi i k \alpha_\ell} + \varepsilon \hat{\lambda}_k^{(\ell)} + \varepsilon^2 \hat{\lambda}_k^{(\ell)} + \mathcal{O}(\varepsilon^2), \quad (11)$$

and

$$\Pi_{k,\varepsilon}^{(\ell)}(\cdot) = \Pi_k^{(\ell)}(\cdot) + \varepsilon \left(\left\langle \cdot, F_k^{(\ell)} \right\rangle \hat{F}_k^{(\ell)} + \left\langle \cdot, \hat{F}_k^{(\ell)} \right\rangle F_k^{(\ell)} \right) + \mathcal{O}(\varepsilon), \quad (12)$$

where $F_k^{(\ell)}(j, x) = f^{(\ell)}(j) e^{2\pi i k x}$ and $\hat{F}_k^{(\ell)}(j, x) := \hat{f}_k(j) e^{2\pi i k x}$. Also, $\langle F_k^{(\ell)}, \hat{F}_k^{(\ell)} \rangle = 0$ and the term $\mathcal{O}(\varepsilon) : L^2(M) \rightarrow L^2(M)$ in (12) is a finite-rank operator such that $\|\mathcal{O}(\varepsilon)\|/\varepsilon \xrightarrow{\varepsilon \rightarrow 0} 0$.

Moreover, explicit formulae for these quantities can be found as follows:

- (1) *If $S = 1$ or $k = 0$:*
 - (a) $\hat{\lambda}_k^{(\ell)}$ and $f_k^{(\ell)}$ are, respectively, the eigenvalues and eigenvectors of $e^{2\pi i k \alpha_1} \dot{W}$;
 - (b) $\hat{\lambda}_k^{(\ell)} = 0$ and $\hat{f}_k^{(\ell)} = 0$; and

- (c) the functions $\mathcal{O}(\varepsilon^2)$ in (11) and $\mathcal{O}(\varepsilon)$ in (12) are identically equal to 0 (defined in Proposition 3.7).
- (2) If $S > 1$ and $k \neq 0$:
- (a) $\hat{\lambda}_k^{(\ell)}$ and $f_k^{(\ell)}$ are, respectively, the eigenvalues and eigenvectors of $\hat{P}_{k,\beta,L}$ (see (8)). Moreover, $\arg(\hat{\lambda}_k^{(\ell)}) = \arg(\lambda_k^{(\ell)}) + \pi$ if $\hat{\lambda}_k^{(\ell)} \neq 0$.
 - (b) $\hat{f}_k^{(\ell)}$ is computed in Theorem 4.5; and
 - (c) $\hat{\hat{\lambda}}_k^{(\ell)}$ is provided in Theorem 5.2.

2.5. Linear response with respect to the band vectors. In this section, we describe the linear response with respect to the rotation speed vector $\alpha \in \mathbb{R}^n$. To emphasise the α dependence on the operator $\mathcal{P}_\varepsilon$ we write it as $\mathcal{P}_{\varepsilon,\alpha}$.

2.5.1. *The case of $\varepsilon > 0$.* For $\varepsilon > 0$ we have a positive result of standard linear response.

Proposition 2.7. Fix $k \in \mathbb{Z}$, $\varepsilon > 0$, and $\alpha \in \mathbb{R}^N$. Assume that $\mathcal{P}_{\varepsilon,\alpha}$ has exactly N distinct eigenvalues $\lambda_{k,\varepsilon,\alpha}^{(\ell)}$, $\ell = 1, \dots, N$ associated with eigenvectors of the form $F_{k,\varepsilon,\alpha}(x, j) = f_{k,\varepsilon,\alpha}(j)e^{2\pi i k x}$. Then, there exists an open neighbourhood A_α of α in $\mathbb{R}^N$ such that the maps

$$U_\alpha \ni \alpha' \mapsto \lambda_{k,\varepsilon,\alpha'}^{(\ell)} \in \mathbb{C} \text{ and } U_\alpha \ni \alpha' \mapsto F_{k,\varepsilon,\alpha'}(\cdot, \cdot) = f_{k,\varepsilon,\alpha'}(\cdot)e^{2\pi i k \cdot} \in \mathcal{C}^0(M, \mathbb{C}), \quad (13)$$

are differentiable. Moreover, one has

$$\frac{\partial}{\partial \alpha} f_{k,\varepsilon,\alpha} = (D_{k,\alpha} W_\varepsilon - \lambda_{k,\varepsilon,\alpha})^{-1} \left(\frac{\partial}{\partial \alpha} D_{k,\alpha} W_\varepsilon - \frac{\partial}{\partial \alpha} \lambda_{k,\varepsilon,\alpha} \right) f_{k,\varepsilon,\alpha}. \quad (14)$$

Proof. See Section 6. □

We remark that by Theorem 2.5(1), for sufficiently small ε one automatically obtains the assumption in Proposition 2.7 that $\mathcal{P}_{\varepsilon,\alpha}$ has exactly N eigenvalues.

2.5.2. *The case of $\varepsilon = 0$.* Let $\Pi_{\varepsilon,\alpha,k}^{(\ell)}$ be the L^2 -orthogonal projection onto $\text{span}\{F_{k,\varepsilon,\alpha}\}$ (defined in (13)). From Theorem 2.5 (2), when $\alpha \in \mathbb{C}^N$ is an N -banded vector we obtain that

$$\Pi_{k,\alpha}^{(\ell)} := \lim_{\varepsilon \rightarrow 0} \Pi_{\varepsilon,\alpha,k}^{(\ell)}$$

is a L^2 -orthogonal projection onto $\text{span}\{F_{k,\alpha}^{(\ell)}\}$, where $F_{k,\alpha}^{(\ell)}(j, x) = \mathbf{1}_{\{\ell\}}(j)e^{2\pi i k x}$. Therefore $\frac{\partial}{\partial \alpha} \Pi_{k,\alpha} = 0$ for every N -banded vector α , since the set of the N -banded vectors is open in $\mathbb{C}^N$. The same type of conclusion can be obtained in the case $k = 0$.

However, when $\varepsilon = 0$, $k \neq 0$, and $\alpha \in \mathbb{C}^N$ is an S -banded vector for some $1 \leq S < N$, the linear response fails. This follows from the fact that the map

$$v \mapsto \Pi_{k,v}^{(\ell)} := \lim_{\varepsilon \rightarrow 0} \Pi_{k,\varepsilon,v}^{(\ell)}$$

is discontinuous in a neighborhood of α . This discontinuity is a consequence of Theorem 2.5 (b) and the denseness of the set of N -banded vectors in $\mathbb{C}^N$.

3. PROOFS OF PROPOSITION 2.2, THEOREM 2.5(a), THEOREM 2.5(2)(a), AND THEOREM 2.6(1)

3.1. Proof of Proposition 2.2. We start by proving results that are valid for any $\alpha \in \mathbb{R}^N$. In future sections, we restrict our analysis to the vector α being an S -band.

3.1.1. *Proof of Proposition 2.2(1).* Lemma 3.1 asserts that eigenfunctions of $\mathcal{P}_\varepsilon$ corresponding to a fixed Fourier index k take the form of a harmonic function defined along each circular fibre, modulated by a function that is solely dependent on the fibres. Recall that $D_{k,\alpha}$ is defined in (4).

Lemma 3.1. *A function $F \in L^2(M)$ is a generalised eigenfunction of $\mathcal{P}_\varepsilon$ with eigenvalue λ if and only if $F(j, x) = f_k(j)e^{2\pi i k x}$ for some $k \in \mathbb{Z}$, where (identifying the function f_k with the vector $(f_k(1), \dots, f_k(N)) \in \mathbb{C}^N$) f_k is a generalised eigenvector of $D_{k,\alpha}W_\varepsilon$ with eigenvalue λ .*

Proof. Given $f \in L^2(M)$, by employing Fourier series, we can express

$$F(j, x) = \sum_{k \in \mathbb{Z}} f_k(j) e^{2\pi i k x}, \text{ where } f_k : \{1, \dots, N\} \rightarrow \mathbb{C}. \quad (15)$$

By (1) and (2) with $\delta = 0$ for every $(j, x) \in M$

$$\begin{aligned} \mathcal{P}_\varepsilon F(j, x) &= \sum_{k \in \mathbb{Z}} \mathcal{P}_\varepsilon (f_k(\cdot) e^{2\pi i k \cdot})(j, x) \\ &= \sum_{k \in \mathbb{Z}} e^{-2\pi i k \alpha_j} (W_\varepsilon f_k)_j e^{2\pi i k x}. \end{aligned} \quad (16)$$

Suppose that $\mathcal{P}_\varepsilon F = \lambda F$. By equating individual Fourier mode coefficients in (16) we obtain $D_{k,\alpha}W_\varepsilon f_k = \lambda f_k$. Similarly, if $D_{k,\alpha}W_\varepsilon f_k = \lambda f_k$ holds, the ansatz (15) provides an eigenfunction F with eigenvalue λ . The case where F is a generalised eigenfunction of F follows similarly. $\square$

3.1.2. *Proof of Proposition 2.2(2).*

Lemma 3.2. *Given $k \in \mathbb{Z}$, let $\{\lambda_{k,\varepsilon}^{(1)}, \dots, \lambda_{k,\varepsilon}^{(N)}\} := \sigma(D_{k,\alpha}W_\varepsilon)$. Then, after properly relabelling the elements of $\sigma(D_{k,\alpha}W_\varepsilon)$, it follows that for every $\varepsilon > 0$ sufficiently small*

$$\left| \lambda_{k,\varepsilon}^{(\ell)} - e^{-2\pi i k \alpha_\ell} \right| \leq 2\varepsilon \text{ for every } \ell \in \{1, \dots, N\}.$$

Proof. For each $j \in \{1, \dots, N\}$, consider the Gershgorin disks

$$\begin{aligned} \Delta_j &:= \left\{ z \in \mathbb{C} : |z - (D_{k,\alpha}W_\varepsilon)_{jj}| \leq \sum_{j' \neq j} |(D_{k,\alpha}W_\varepsilon)_{jj'}| \right\} \\ &\subset \{z \in \mathbb{C} : |z - e^{-2\pi i k \alpha_j}| \leq 2\kappa\varepsilon\} =: \tilde{\Delta}_j, \end{aligned}$$

where $\kappa = \max_{1 \leq j \leq N} |\dot{W}_{jj}|$.

Take $\varepsilon_0 > 0$ sufficiently small such that the number of connected components of $\bigcup_{j=1}^N \tilde{\Delta}_j$ is constant for every $0 < \varepsilon < \varepsilon_0$. From the Gershgorin circle theorem [14, Theorem 6.11] we obtain that $\left| \lambda_{k,\varepsilon}^{(j)} - e^{-2\pi i k \alpha_j} \right| \leq 2\kappa\varepsilon$ for every $j \in \{1, \dots, N\}$. $\square$

3.2. Distinct eigenvalues of $\mathcal{P}_\varepsilon$ for small $\varepsilon > 0$. The proofs of Theorem 2.5(1), Theorem 2.5(2)(i), and Theorem 2.6(1) all rely on the eigenspaces of $\mathcal{P}_\varepsilon$ being one-dimensional. Recalling the correspondence between the spectrum of $\mathcal{P}_\varepsilon$ and $D_{k,\alpha}W_\varepsilon$ established in Lemma 3.1, we adopt the following simplified notation in this and all subsequent sections. Fixing β , L , and $k \in \mathbb{Z}$, for every $\varepsilon > 0$ we write

$$D_\beta := D_{k,\beta,L} \text{ and } P_{\varepsilon,\beta} := D_\beta W_\varepsilon. \quad (17)$$

For $v, w \in \mathbb{C}^N$ we denote the inner product between v and w as $\langle v, w \rangle = \sum_{j=1}^N v_j \overline{w_j}$, and for every $s \in \{1, \dots, N\}$ we set

$$\begin{aligned} V_s &:= \{v \in \mathbb{C}^N : v_j = 0 \text{ if } j \leq N_{s-1} \text{ or } j > N_s\} \\ &= \{v \in \mathbb{C}^N : v_j = 0 \text{ for every } j \in \{1, \dots, N\} \setminus B_s\}. \end{aligned} \quad (18)$$

The following three results establish that for every $\beta \in \Gamma$ the matrix $P_{\varepsilon, \beta}$ is diagonalizable for every $\varepsilon > 0$ sufficiently small.

Theorem 3.3. *Under the hypotheses of Theorem 2.5 there exists $\gamma_k > 0$ such that for every $0 < \varepsilon < \gamma_k$ the matrix $P_{\varepsilon, \beta}$ has N distinct eigenvalues.*

Before proving the above theorem, we establish three preparatory lemmas.

Lemma 3.4. *Fix $s \in \{1, \dots, S\}$. For each $\varepsilon > 0$ choose a sequence of unit-norm vectors $\{f_\varepsilon\}_{\varepsilon > 0} \subset \mathbb{C}^N$ such that for every $\varepsilon > 0$,*

$$P_{\varepsilon, \beta} f_\varepsilon = \lambda_\varepsilon f_\varepsilon \text{ and } \lambda_\varepsilon \xrightarrow{\varepsilon \rightarrow 0} e^{-2\pi i k \beta_s}.$$

Then, $(f_\varepsilon)_j \xrightarrow{\varepsilon \rightarrow 0} 0$ for every $j \in \{1, \dots, N\} \setminus B_s$.

Proof. Since $\|f_\varepsilon\| = 1$ and for every ε small enough there exists a converging subsequence $\{f_{\varepsilon_r}\}_{r \in \mathbb{N}} \subset \{f_\varepsilon\}_{\varepsilon > 0}$ such that $f_{\varepsilon_r} \xrightarrow{\varepsilon_r \rightarrow 0} f_*$. It follows that

$$D_\beta f_* = \lim_{k \rightarrow \infty} D_\beta W_{\varepsilon_r} f_{\varepsilon_r} = \lambda_{\varepsilon_r} f_{\varepsilon_r} = e^{-2\pi i k \beta_s} f_*.$$

Since $\beta \in \Gamma$, then $\min_{s_1 \neq s_2} |e^{-2\pi i k \beta_{s_1}} - e^{-2\pi i k \beta_{s_2}}| > 0$ which implies that $f_* = 0$ for every $j \in \{1, \dots, N\} \setminus B_s$. Hence, for every subsequence of $\{f_\varepsilon\}_{\varepsilon > 0}$, it is possible to construct a subsubsequence such that all coordinates $j \in \{1, \dots, N\} \setminus B_s$ converge to 0 as $\varepsilon \rightarrow 0$, which implies the desired result. $\square$

Lemma 3.5. *Assume that there exist vectors $f, g \in \mathbb{C}^N$, $\beta \in \mathbb{R}^S$ and $\varepsilon > 0$ such that*

$$P_{\varepsilon, \beta} f = \lambda f \text{ and } P_{\varepsilon, \beta} g = \lambda g + f.$$

Then, $\langle f, D_\beta \bar{f} \rangle = 0$.

Proof. Since $P_{\varepsilon, \beta} g = \lambda g + f$, then $\overline{D_\beta W_\varepsilon g} = \overline{\lambda g + f}$. Therefore $W_\varepsilon \bar{g} = \bar{\lambda} D_\beta \bar{g} + D_\beta \bar{f}$. Hence,

$$\begin{aligned} \langle f, D_\beta \bar{g} \rangle &= \frac{1}{\lambda} \langle P_{\varepsilon, \beta} f, D_\beta \bar{g} \rangle = \frac{1}{\lambda} \langle D_\beta W_\varepsilon f, D_\beta \bar{g} \rangle = \frac{1}{\lambda} \langle f, W_\varepsilon \bar{g} \rangle \\ &= \frac{1}{\lambda} \langle f, \bar{\lambda} D_\beta \bar{g} + D_\beta \bar{f} \rangle = \langle f, D_\beta \bar{g} \rangle + \frac{1}{\lambda} \langle f, D_\beta \bar{f} \rangle. \end{aligned}$$

Thus, $\langle f, D_\beta \bar{f} \rangle = 0$. $\square$

Lemma 3.6. *Let $\beta \in \mathbb{R}^S$ and $\{f_\varepsilon\}_{\varepsilon > 0} \subset \mathbb{C}^N$ be a sequence of unit-norm vectors such that for every $\varepsilon > 0$ small*

$$P_{\varepsilon, \beta} f_\varepsilon = \lambda_\varepsilon f_\varepsilon \text{ and } \lambda_\varepsilon \xrightarrow{\varepsilon \rightarrow 0} e^{-2\pi i k \beta_s} \text{ for some } s \in \{1, \dots, S\}.$$

Then, the accumulation points of $\{f_\varepsilon\}_{\varepsilon > 0}$ as $\varepsilon \rightarrow 0$ are eigenvectors of the matrix $\hat{P}_{k, \beta, L}$ (defined in 8).

Proof. Consider a converging subsequence $\{f_{\varepsilon_r}\}_{r \in \mathbb{N}} \subset \{f_\varepsilon\}_{\varepsilon > 0}$ such that $f_{\varepsilon_r} \xrightarrow{\varepsilon_r \rightarrow 0} f_*$. From Lemma 3.2, we can choose a subsubsequence $\{\varepsilon'_r\}_{r' \in \mathbb{N}} \subset \{\varepsilon_r\}_{r \in \mathbb{N}}$ such that

$$\frac{\lambda_{\varepsilon'_r} - e^{-2\pi i k \beta_s}}{\varepsilon'_r} \xrightarrow{\varepsilon'_r \rightarrow 0} \lambda_*.$$

Moreover, from Lemma 3.4 we have that $f_* \in V_s$ (see (18)). Since

$$\lambda_\varepsilon f_\varepsilon = P_{\varepsilon,\beta} f_\varepsilon = D_\beta W_\varepsilon f_\varepsilon = D_\beta (\text{Id} + \varepsilon \dot{W}) f_\varepsilon.$$

We obtain that $\dot{W} f_\varepsilon = \frac{1}{\varepsilon} (\lambda_\varepsilon D_{k,-\beta,L} f_\varepsilon - f_\varepsilon)$ for each $\varepsilon > 0$. Let $\pi_s : \mathbb{C}^N \rightarrow V_s$ the orthogonal projection from $\mathbb{C}^N$ into V_s . It follows that

$$\begin{aligned} \pi_s \dot{W} f_\varepsilon &= \frac{1}{\varepsilon} \pi_s (\lambda_\varepsilon D_{k,-\beta,L} f_\varepsilon - f_\varepsilon) = \frac{1}{\varepsilon} \pi_s (\lambda_\varepsilon D_{k,-\beta,L} f_\varepsilon - f_\varepsilon) \\ &= \frac{e^{2\pi i k \beta_s} \lambda_\varepsilon - 1}{\varepsilon} \pi_s f_\varepsilon. \end{aligned}$$

By taking $\varepsilon'_r \rightarrow 0$ and observing that $\pi_s f_* = f_*$, we obtain that $\pi_s \dot{W} f_* = \lambda_* f_*$. Thus, $\hat{W}_L f_* = \lambda_* f_*$ (see (9)). Therefore, we obtain that

$$\hat{P}_{k,\beta,L} f_* = D_\beta \hat{W}_L f_* = e^{2\pi i k \beta_s} \lambda_* f_*.$$

□

Proof of Theorem 3.3. Assume for a contradiction that for every $\varepsilon_0 > 0$ there exists $0 < \varepsilon < \varepsilon_0$ such that $D_\beta W_\varepsilon$ does not have N distinct eigenvalues. Then, there exists a sequence of real numbers $\varepsilon_n \rightarrow 0$, as $n \rightarrow \infty$, and sequences of vectors $\{f_{\varepsilon_n}\}_{n \in \mathbb{N}}, \{g_{\varepsilon_n}\}_{n \in \mathbb{N}} \subset \mathbb{C}^N$ such that for every $n \in \mathbb{N}$ one has $\|f_{\varepsilon_n}\| = 1$ and either

- (1) $P_{\varepsilon,\beta} g_{\varepsilon_n} = \lambda_\varepsilon g_{\varepsilon_n} + f_{\varepsilon_n}$, or
- (2) f_{ε_n} and g_{ε_n} are linearly independent and

$$P_{\varepsilon_n,\beta} f_{\varepsilon_n} = \lambda_{\varepsilon_n} f_{\varepsilon_n} \text{ and } P_{\varepsilon_n,\beta} g_{\varepsilon_n} = \lambda_{\varepsilon_n} g_{\varepsilon_n}.$$

We first consider scenario (1). By passing through a subsequence, if necessary, we can assume that there exists f_* and λ_* such that

$$f_{\varepsilon_n} \xrightarrow{n \rightarrow \infty} f_* \text{ and } \lambda_{\varepsilon_n} \xrightarrow{n \rightarrow \infty} e^{-2\pi i k \beta_s} \text{ for some } s \in \{1, \dots, S\}.$$

From Lemmas 3.4, 3.5 and 3.6 respectively, we obtain that

- (a) $f_* \in V_s$ (see (18)) for some $s \in \{1, \dots, S\}$;
- (b) $\langle f_*, D_\beta \bar{f}_* \rangle = \lim_{n \rightarrow \infty} \langle f_{\varepsilon_n}, D_\beta \bar{f}_{\varepsilon_n} \rangle = 0$;
- (c) f_* is an eigenvector of $\hat{P}_{k,\beta,L}$.

From the block structure of $\hat{P}_{k,\beta,L}$ we have that f_* that is an eigenvector of $\hat{W}_L$ (see (9)). Since $f_* \in V_s$ and all the blocks of $\hat{W}_L$ are symmetric real matrices with distinct (real) eigenvalues, we obtain that f_* and $\bar{f}_*$ are linearly dependent, which implies that $\langle f_*, D_\beta \bar{f}_* \rangle \neq 0$, contradicting item (b) above.

Now we assume (2). Define

$$h_{\varepsilon_n} := \frac{1}{\|g_{\varepsilon_n} - \langle g_{\varepsilon_n}, f_{\varepsilon_n} \rangle f_{\varepsilon_n}\|} (g_{\varepsilon_n} - \langle g_{\varepsilon_n}, f_{\varepsilon_n} \rangle f_{\varepsilon_n}).$$

By passing, if necessary, through a subsequence, we can assume $f_*, h_* \in \mathbb{C}^N$ that

$$f_{\varepsilon_n} \xrightarrow{n \rightarrow \infty} f_*, \quad h_{\varepsilon_n} \xrightarrow{n \rightarrow \infty} h_* \text{ and } \frac{\lambda_{\varepsilon_n} - e^{-2\pi i k \beta_s}}{\varepsilon_n} \xrightarrow{n \rightarrow \infty} \lambda_* \text{ for some } s \in \{1, \dots, S\}.$$

From Lemma 3.4 we have that $f_*, h_* \in V_s \setminus \{0\}$ and by construction $\langle f_*, h_* \rangle = 0$. From Lemma 3.6 we obtain that

$$\hat{P}_{k,\beta,L} f_* = \lambda_* f_* \text{ and } \hat{P}_{k,\beta,L} h_* = \lambda_* h_*,$$

and therefore

$$\pi_s \hat{W}_L f_* = e^{2\pi k \beta_s} \lambda_* f_* \text{ and } \pi_s \hat{W}_L h_* = e^{2\pi k \beta_s} \lambda_* h_*, \quad (19)$$

where $\pi_s : \mathbb{C}^N \rightarrow V_s$ is the $\mathbb{C}^N$ -orthogonal projection onto V_s . Since W_ε is an L -admissible family of matrices, Definition 2.4(3) implies that $\pi_s \hat{W}_L : V_s \rightarrow V_s$ has N distinct eigenvalues, which contradict (19) since $\langle f_*, h_* \rangle = 0$. $\square$

3.3. Proof of Theorem 2.5(2)(a) and Theorem 2.6(1).

Proposition 3.7. *Under the assumptions and notation of Theorem 2.5, if either $S = 1$ or $k = 0$, then for each $\ell \in \{1, \dots, N\}$,*

$$\lambda_{k,\varepsilon}^{(\ell)} = e^{-2\pi i k \beta_1} + \varepsilon \hat{\lambda}_k^{(\ell)} \text{ and } F_{k,\varepsilon}^{(\ell)}(j, x) = f_k^{(\ell)}(j) e^{2\pi i k x}.$$

Moreover, $\{f_k^{(1)}, \dots, f_k^{(N)}\}$ is an orthonormal basis of $e^{-2\pi i k \alpha_1} \dot{W}$, and for each $\ell \in \{1, \dots, N\}$, $\hat{\lambda}_k^{(\ell)}$ is an eigenvalue of $e^{-2\pi i k \alpha_1} \dot{W}$ associated with $f_k^{(\ell)}$. In particular, $\Pi_{k,\varepsilon}^{(\ell)} = \Pi_k^{(\ell)}$ for every $\varepsilon > 0$, where $\Pi_k^{(\ell)}$ is the L^2 -orthogonal projection onto $\text{span}\{F_k^{(\ell)}\}$, where $F(j, x) = f_k^{(\ell)}(j) e^{2\pi i k x}$.

Proof. From Lemma 3.1 it follows that $\mathcal{P}_\varepsilon F_\varepsilon = \lambda_\varepsilon F_\varepsilon$ if and only if $F(j, x) = f(j) e^{2\pi i k x}$ and $D_\beta W_\varepsilon f_\varepsilon = \lambda_\varepsilon f_\varepsilon$. Since either $S = 1$ or $k = 0$ we obtain that $D_\beta = e^{-2\pi i k \beta_1} \text{Id}_N$. Therefore

$$D_\beta W_\varepsilon = e^{-2\pi i k \beta_1} W_\varepsilon = e^{-2\pi i k \beta_1} (\text{Id}_N + \varepsilon \dot{W}).$$

Since either $S = 1$ or $k = 0$ we obtain that $e^{-2\pi i k \beta_1} (\text{Id}_N + \varepsilon \dot{W}) f_\varepsilon = \lambda_\varepsilon f_\varepsilon$ if and only if

$$e^{-2\pi i k \alpha_1} \dot{W} f_\varepsilon = \frac{\lambda_\varepsilon - e^{-2\pi i k \beta_1}}{\varepsilon} f_\varepsilon.$$

Hence, for every $\ell \in \{1, \dots, N\}$

$$\lambda_{k,\varepsilon}^{(\ell)} = e^{-2\pi i k \beta_1} + \varepsilon \hat{\lambda}_k^{(\ell)} \text{ and } f_{k,\varepsilon}^{(\ell)} = f_k^{(\ell)}.$$

Observe that $f_k^{(\ell)}$ is an orthonormal basis since W_ε is an L -admissible family of matrices. $\square$

4. PROOF OF THEOREM 2.5(2)(b) AND THEOREM 2.6: THE LINEAR COMPONENT OF (11) AND PART(2)(a)

To simplify notation we fix from now on $N, S \in \mathbb{N}$ with $1 < S \leq N$, a cluster length vector $L \in \mathbb{R}^S$, a family of L -admissible family of matrices $W_\varepsilon = \text{Id} + \varepsilon \dot{W}$, and $k \in \mathbb{N}$. Moreover, given a function $h : \{1, \dots, N\} \rightarrow \mathbb{C}$, we denote $(h)_j := h(j)$ for every $j \in \{1, \dots, N\}$, and we make no distinction between the vector $((h)_1, \dots, (h)_N)$ and the function h . Finally, we say $(h)_j = 0$ if $h(j)$ is undefined, i.e. when $j \in \mathbb{Z} \setminus \{1, \dots, N\}$.

Below, we recall the definition of the complex projective space $\mathbb{CP}^{N-1}$ which is used throughout the rest of the proofs of Theorems 2.5 and 2.6.

Definition 4.1 (Complex Projective Space). Given $N \in \mathbb{N}$, we denote

$$\mathbb{CP}^{N-1} = \mathbb{C}^N \setminus \{0\} / \sim,$$

where $\sim$ is the equivalence relation in $\mathbb{C}^N \setminus \{0\}$

$$v \sim w \text{ if and only if } v = \lambda w \text{ for some } \lambda \in \mathbb{C}.$$

We induce a differential structure in $\mathbb{CP}^N$ via the quotient map

$$\pi : \mathbb{C}^n \setminus \{0\} \rightarrow \mathbb{CP}^N$$

$$v \mapsto [v]$$

where $[v] \in \mathbb{C}^n$ is the equivalence class of v in $\sim$ (see [19, Chapter 1, page 31]). Given $w, v \in \mathbb{CP}^{N-1}$ we define the distance between w and v in $\mathbb{CP}^{N-1}$ as

$$\text{dist}_{\mathbb{CP}^{N-1}}(v, w) = \min_{\theta \in [0, 2\pi)} \left\| \frac{v}{\|v\|} - e^{i\theta} \frac{w}{\|w\|} \right\|.$$

Theorem 4.2. *Let $\beta \in \Gamma$, for each $\varepsilon > 0$ small enough, let $\{f_\varepsilon^{(\ell)}\}_{\ell=1}^N \subset \mathbb{C}^N$ denote a unit-norm eigenbasis of $P_{\varepsilon, \beta}$. Then,*

(i) *there exists an orthonormal eigenbasis $\{f^{(1)}, \dots, f^{(N)}\} \subset \mathbb{C}^N$ of*

$$\hat{P}_\beta := D_\beta \hat{W}_L,$$

where $\hat{W}_L$ is defined in (9) and D_β in (17), such that for every $j \in \{1, \dots, N\}$, one has

$$[f_\varepsilon^{(\ell)}] \xrightarrow{\varepsilon \rightarrow 0} [f^{(\ell)}] \text{ in } \mathbb{CP}^{N-1} \text{ for every } \ell \in \{1, \dots, N\}. \quad (20)$$

(ii) *For $s \in \{1, \dots, N\}$ and $\ell \in B_s$, let $\hat{\lambda}^{(\ell)}$ be the eigenvalue associated to the eigenvector $f^{(\ell)}$ of $\hat{P}_\beta$. Then*

$$\lambda_\varepsilon^{(\ell)} = e^{-2\pi i k \beta_s} + \varepsilon \hat{\lambda}^{(\ell)} + o(\varepsilon). \quad (21)$$

Moreover, $\arg(\hat{\lambda}^{(\ell)}) = \arg(\lambda^{(\ell)}) + \pi$ if $|\hat{\lambda}^{(\ell)}| > 0$.

4.1. Proof of Theorem 4.2. The proof of Theorem 4.2 is lengthy. In what follows, we establish two preparatory lemmas before presenting its proof. For every $\varepsilon > 0$ small enough we write $\{f_\varepsilon^{(\ell)}\}_{\ell=1}^N \subset \mathbb{C}^N$ as a unit-norm eigenbasis of $P_{\varepsilon, \beta}$, and by appealing to standard matrix perturbation theory [15, Chapter Two §1] we obtain that for every $\ell \in \{1, \dots, N\}$, the map

$$\varepsilon \in (0, \gamma_k) \mapsto f_\varepsilon^{(\ell)} \in \mathbb{C}^N$$

is $\mathcal{C}^\infty$. We define $\lambda_\varepsilon^{(\ell)}$ as the eigenvalue associated to the eigenvector $f_\varepsilon^{(\ell)}$. Recall that

$$\lambda_\varepsilon^{(\ell)} \rightarrow e^{-2\pi i k \beta_s} \text{ if } \ell \in B_s \text{ as } \varepsilon \rightarrow 0.$$

We mention that standard perturbation theory asserts [15, Chapter Two §1] that the above map $\varepsilon \in (0, \gamma_k) \mapsto \lambda_\varepsilon^{(\ell)} \in \mathbb{C}$ is $\mathcal{C}^\infty$. Finally, for every $\ell \in \{1, \dots, N\}$, $\Pi_\varepsilon^{(\ell)} : L^2(M) \rightarrow \text{span}\{F_\varepsilon^{(\ell)}\}$ denotes the L^2 -orthogonal projection onto the $\text{span}\{F_\varepsilon^{(\ell)}\}$ where $F_\varepsilon(j, x) := f_\varepsilon^{(\ell)} e^{2\pi i k x}$.

Lemma 4.3. *Let $1 \leq \ell \neq m \leq N$, then $\langle f_\varepsilon^{(\ell)}, D_\beta \overline{f_\varepsilon^{(m)}} \rangle = 0$ for every $\ell \neq m$ and $\varepsilon > 0$ small.*

Proof. From Lemma 3.2 $\lambda_\varepsilon^{(\ell)} \neq 0$ for every $\ell \in \{1, \dots, N\}$ and $\varepsilon > 0$ sufficiently small. Observe that

$$\begin{aligned} \langle f_\varepsilon^{(\ell)}, D_\beta \overline{f_\varepsilon^{(m)}} \rangle &= \frac{1}{\lambda_\varepsilon^{(\ell)}} \langle P_{\varepsilon, \beta} f_\varepsilon^{(\ell)}, D_\beta \overline{f_\varepsilon^{(m)}} \rangle = \frac{1}{\lambda_\varepsilon^{(\ell)}} \langle D_\beta W_\varepsilon f_\varepsilon^{(\ell)}, D_\beta \overline{f_\varepsilon^{(m)}} \rangle \\ &= \frac{1}{\lambda_\varepsilon^{(\ell)}} \langle W_\varepsilon f_\varepsilon^{(\ell)}, \overline{f_\varepsilon^{(m)}} \rangle = \frac{1}{\lambda_\varepsilon^{(\ell)}} \langle f_\varepsilon^{(\ell)}, W_\varepsilon \overline{f_\varepsilon^{(m)}} \rangle. \end{aligned} \quad (22)$$

Since $D_\beta W_\varepsilon f_\varepsilon^{(m)} = \lambda_\varepsilon^{(m)} f_\varepsilon^{(m)}$, it follows that $W_\varepsilon f_\varepsilon^{(m)} = \lambda_\varepsilon^{(m)} D_{-\beta} f_\varepsilon^{(m)}$ and therefore $W_\varepsilon \overline{f_\varepsilon^{(m)}} = \overline{\lambda_\varepsilon^{(m)}} D_\beta \overline{f_\varepsilon^{(m)}}$. Combining the last equality with (22), we obtain

$$\langle f_\varepsilon^{(\ell)}, D_\beta \overline{f_\varepsilon^{(m)}} \rangle = \frac{1}{\lambda_\varepsilon^{(\ell)}} \langle f_\varepsilon^{(\ell)}, W_\varepsilon \overline{f_\varepsilon^{(m)}} \rangle = \frac{1}{\lambda_\varepsilon^{(\ell)}} \langle f_\varepsilon^{(\ell)}, \overline{\lambda_\varepsilon^{(m)}} D_\beta \overline{f_\varepsilon^{(m)}} \rangle = \frac{\lambda_\varepsilon^{(m)}}{\lambda_\varepsilon^{(\ell)}} \langle f_\varepsilon^{(\ell)}, D_\beta \overline{f_\varepsilon^{(m)}} \rangle.$$

Finally, from Theorem 3.3 we obtain that $\lambda_\varepsilon^{(m)} \neq \lambda_\varepsilon^{(\ell)}$ when $\varepsilon > 0$ is small enough. Thus, $\langle f_\varepsilon^{(\ell)}, D_\beta \overline{f_\varepsilon^{(m)}} \rangle = 0$. $\square$

Lemma 4.4. *For every $1 \leq \ell \neq m \leq N$, $\lim_{\varepsilon \rightarrow 0} \langle f_\varepsilon^{(\ell)}, f_\varepsilon^{(m)} \rangle = 0$.*

Proof. Let $s_\ell, s_m \in \{1, \dots, S\}$ such that $\ell \in B_{s_\ell}$ and $m \in B_{s_m}$.

Consider a converging subsequence of $\{\langle f_{\varepsilon_n}^{(\ell)}, f_{\varepsilon_n}^{(m)} \rangle\}_{n \in \mathbb{N}} \subset \{\langle f_\varepsilon^{(\ell)}, f_\varepsilon^{(m)} \rangle\}_{\varepsilon > 0}$. By passing through a subsequence, if necessary, there exist complex vectors $f_*^{(\ell)}, f_*^{(m)} \in \mathbb{C}^N$

$$\lim_{n \rightarrow \infty} f_{\varepsilon_n}^{(\ell)} = f_*^{(\ell)} \text{ and } \lim_{n \rightarrow \infty} f_{\varepsilon_n}^{(m)} = f_*^{(m)}.$$

From Lemma 3.6, we have that $f_*^{(m)}$ and $f_*^{(\ell)}$ are eigenvectors of the matrix $\hat{P}_\beta$ (see 4.2). Moreover, Lemma 3.4 implies that $f_*^{(\ell)} \in V_{s_\ell}$ and $f_*^{(m)} \in V_{s_m}$. Therefore, $f_*^{(\ell)}$ and $f_*^{(m)}$ are eigenvectors of the real symmetric matrix $\hat{W}_L$. Hence, $\overline{f_*^{(m)}} = a f_*^{(m)}$ for some $a \in \mathbb{C}$. Finally, using Lemma 4.3 we obtain that for every $m \neq \ell$

$$\begin{aligned} 0 &= \lim_{n \rightarrow \infty} \frac{e^{-2\pi i k \beta_{s_m}}}{\bar{a}} \langle f_{\varepsilon_n}^{(\ell)}, D_\beta \overline{f_{\varepsilon_n}^{(m)}} \rangle = \frac{e^{-2\pi i k \beta_{s_m}}}{\bar{a}} \langle f_*^{(\ell)}, D_\beta \overline{f_*^{(m)}} \rangle \\ &= \langle f_*^{(\ell)}, f_*^{(m)} \rangle = \lim_{n \rightarrow \infty} \langle f_{\varepsilon_n}^{(\ell)}, f_{\varepsilon_n}^{(m)} \rangle \end{aligned}$$

which finishes the proof of the lemma. $\square$

With the above results established, we can finally prove Theorem 4.2.

Proof of Theorem 4.2. We start by establishing (21), which implies the first part of (ii). Let $\ell \in \{1, \dots, N\}$, from Lemma 3.2 then there exists $s \in \{1, \dots, S\}$, such that $\lambda_\varepsilon^{(\ell)} \rightarrow e^{-2\pi i k \beta_s}$. Define $\pi_s : \mathbb{C}^N \rightarrow V_s$ as the $\mathbb{C}^N$ -orthogonal projection onto V_s (cf. (18)). Observe that

$$\begin{aligned} \hat{P}_\beta f_\varepsilon^{(\ell)} &= D_\beta \hat{W}_L f_\varepsilon^{(\ell)} D_\beta \hat{W}_L \pi_s f_\varepsilon^{(\ell)} + D_\beta \hat{W}_L (f_\varepsilon^{(\ell)} - \pi_s f_\varepsilon^{(\ell)}) \\ &= \frac{1}{\varepsilon} (\lambda_\varepsilon^{(\ell)} - e^{-2\pi i k \beta_s}) \pi_s f_\varepsilon^{(\ell)} + \mathcal{O} \left(\max_{j \in \{1, \dots, N\} \setminus B_s} |f_\varepsilon^{(\ell)}(j)| \right) \end{aligned} \quad (23)$$

Assume for a contradiction that the sequence

$$\left\{ \frac{1}{\varepsilon} (\lambda_\varepsilon^{(\ell)} - e^{-2\pi i k \beta_s}) \right\}_{\varepsilon > 0} \quad (24)$$

does not converge as $\varepsilon \rightarrow 0$. From Lemma 3.2, (24) is bounded for $\varepsilon > 0$ small enough. Since (24) is bounded and we are assuming that the limit of such a sequence does not exist, there exist sequences of real numbers $\{\varepsilon_n\}_{n \in \mathbb{N}}$ and $\{\varepsilon'_n\}_{n \in \mathbb{N}}$ and complex numbers $\lambda_{(1)} \neq \lambda_{(2)} \in \mathbb{C}$ such that:

- (1) $\varepsilon_n, \varepsilon'_n \xrightarrow{n \rightarrow \infty} 0$;
- (2) $\varepsilon_n \leq \varepsilon'_n \leq \varepsilon_{n+1}$ for every $n \in \mathbb{N}$; and
- (3) $\frac{1}{\varepsilon_n} (\lambda_{\varepsilon_n}^{(\ell)} - e^{-2\pi i k \beta_s}) \xrightarrow{n \rightarrow \infty} \lambda_{(1)}$ and $\frac{1}{\varepsilon'_n} (\lambda_{\varepsilon'_n}^{(\ell)} - e^{-2\pi i k \beta_s}) \rightarrow \lambda_{(2)}$.

Using that $\varepsilon \in (0, \gamma_k) \mapsto \lambda_\varepsilon^{(\ell)} \in \mathbb{C}$ is continuous we can construct a sequence of real numbers $\{\varepsilon''_r\}_{r'' \in \mathbb{N}}$ such that

- (1) $\varepsilon_n \leq \varepsilon''_n \leq \varepsilon'_n$ for every $n \in \mathbb{N}$;
- (2) $\frac{1}{\varepsilon''_n} (\lambda_{\varepsilon''_n}^{(\ell)} - e^{-2\pi i k \beta_s}) \xrightarrow{n \rightarrow \infty} \lambda_{(3)} \notin \sigma(D_\beta \hat{W}_L)$; and
- (3) $f_{\varepsilon''_n}^{(\ell)} \rightarrow f^* \in V_s$ with $\|f^*\|_2 = 1$.

Taking the limit $\varepsilon_r'' \rightarrow 0$ in (23) we obtain that

$$\begin{aligned} D_\beta \hat{W}_L f^* &= \hat{P}_\beta f^* = \lim_{\varepsilon_r'' \rightarrow 0} \hat{P}_\beta f_{\varepsilon_r''}^{(\ell)} \\ &= \lim_{\varepsilon_r'' \rightarrow 0} \left[\frac{1}{\varepsilon_r''} \left(\lambda_{\varepsilon_r''}^{(\ell)} - e^{-2\pi i k \beta_s} \right) \pi_s f_{\varepsilon_r''}^{(\ell)} + \mathcal{O} \left(\max_{j \in \{1, \dots, N\} \setminus B_s} |f_{\varepsilon_r''}^{(\ell)}(j)| \right) \right] \\ &= \lambda_{(3)} f^* + 0 = \lambda_{(3)} f^*. \end{aligned}$$

Since $\lambda_{(3)} \notin \sigma(D_\beta \hat{W}_L)$ this yields a contradiction, therefore $\left\{ \frac{1}{\varepsilon} \left(\lambda_\varepsilon^{(\ell)} - e^{-2\pi i k \beta_s} \right) \right\}_{\varepsilon > 0}$ must converge to an eigenvalue $\hat{\lambda}^{(\ell)}$ of $D_\beta \hat{W}_L$. We thus conclude that (21) holds.

In the following we show (i). Recall that from the assumption imposed on $\hat{W}_L$ (see Definition 2.4 (3)) that $\hat{P}_\beta = D_\beta \hat{W}_L$ admits $L_s = \#B_s$ distinct eigenvalues lying in V_s and that $\ell \in B_s$. From (23) and Lemma 3.4 it is clear that all the accumulation points of $\{f_\varepsilon^{(\ell)}\}_{\varepsilon > 0}$ lie in the one-dimensional complex vector space $\ker(\hat{P}_\beta - \hat{\lambda}^{(\ell)}) \cap V_s$. Choosing $f^{(\ell)} \in V_s \setminus \{0\}$ such that $\hat{P}_\beta f^{(\ell)} = \hat{\lambda}^{(\ell)} f^{(\ell)}$ we have that $[f_\varepsilon^{(\ell)}] \xrightarrow{\varepsilon \rightarrow 0} [f^{(\ell)}]$ in $\mathbb{CP}^{N-1}$. Since $\hat{P}_\beta$ has $\#B_s$ different eigenvectors lying in V_s , using Lemma 4.4 we obtain that $\{f^{(\ell')}\}_{\ell' \in B_s}$ is an orthonormal basis of V_s . Observing that $\mathbb{C}^N = V_1 \oplus \dots \oplus V_S$ and repeating the same argumentation presented in this paragraph for every $s' \in \{1, \dots, S\}$ we obtain that $\{f^{(1)}, \dots, f^{(N)}\}$ is an orthonormal basis of $\mathbb{C}^N$, which concludes the proof of (i).

Finally, we show the second part of (ii), i.e. we show that $\arg(\hat{\lambda}^{(\ell)}) = \arg(\lambda^{(\ell)}) + \pi$ for every $\ell \in \{1, \dots, N\}$ if $\lambda^{(\ell)} \neq 0$. Since $\hat{P}_\beta = D_\beta \hat{W}_L$ is a block matrix, there exists $s \in \{1, \dots, S\}$ such that $\lambda^{(\ell)}$ is also an eigenvalue of the $L_s \times L_s$ matrix $e^{-2\pi i k \beta_s} \hat{W}_s$. In this way, $\lambda^{(\ell)} = e^{-2\pi i k \beta_s} \rho_s$ where $\rho_s \in \mathbb{R}$ is an eigenvalue of the real symmetric matrix $\hat{W}_s$. From Definition 2.4 (1), all the diagonal entries of $\hat{W}_s$ are negative real numbers and the sum of each row is non-positive. As a consequence of the Gershgorin circle theorem [14, Theorem 6.11], we obtain that all eigenvalues of $\hat{W}_s$ are non-positive, i.e. $\rho_s \leq 0$. Hence, the complex argument of $\hat{\lambda}^{(\ell)} = e^{-2\pi i k \beta_s} \rho_s$ is equal to $-2\pi k \beta_s + \pi \pmod{2\pi}$, as long as $\hat{\lambda}^{(\ell)} \neq 0$. $\square$

The following theorem discusses the linear response of the eigenprojections $\Pi_{k,\varepsilon}^{(\ell)}$.

Theorem 4.5. *Let $\ell \in \{1, \dots, N\}$, $s_\ell \in \{1, \dots, S\}$ such that $\ell \in B_{s_\ell}$ and $\beta \in \Gamma$. For each $\varepsilon > 0$ small enough, consider $\{f_\varepsilon^{(1)}, \dots, f_\varepsilon^{(N)}\}$ and $\{f^{(1)}, \dots, f^{(N)}\}$ as in Theorem 4.2. Then,*

(i) *there exist vectors $\hat{f}^{(1)}, \dots, \hat{f}^{(N)} \in \mathbb{C}^N$, so that*

$$[f_\varepsilon^{(\ell)}] = [f^{(\ell)} + \varepsilon \hat{f}^{(\ell)} + o(\varepsilon)] \text{ for every } \ell \in \{1, \dots, N\},$$

where

$$\begin{aligned} \hat{f}^{(\ell)} &= \sum_{s=1, s \neq s_\ell}^S \sum_{r \in B_{s_\ell} \setminus \{\ell\}} \frac{1}{\hat{\lambda}^{(\ell)} - \hat{\lambda}^{(r)}} \frac{1}{e^{-2\pi i k \beta_s} - e^{-2\pi i k \beta_{s_\ell}}} \left\langle \pi_s D_\beta \dot{W} f^{(\ell)}, \pi_s \dot{W} D_{-\beta} f^{(r)} \right\rangle f^{(r)} \\ &+ \sum_{s=1, s \neq s_\ell}^S \sum_{r \in B_s} \frac{1}{e^{-2\pi i k \beta_s} - e^{-2\pi i k \beta_{s_\ell}}} \left\langle D_\beta \dot{W} f^{(\ell)}, f^{(r)} \right\rangle f^{(r)}. \end{aligned} \quad (25)$$

and $\pi_s : \mathbb{C}^N \rightarrow V_s$ is the $\mathbb{C}^N$ -orthogonal projection onto V_s .

(ii) *Moreover,*

$$\Pi_\varepsilon^{(\ell)} = \Pi^{(\ell)} + \varepsilon \left(\left\langle \cdot, F^{(\ell)} \right\rangle \hat{F}^{(\ell)} + \left\langle \cdot, \hat{F}^{(\ell)} \right\rangle F^{(\ell)} \right) + o(\varepsilon),$$

where $F^{(\ell)}(j, x) = f^{(\ell)}(j)e^{2\pi i k x}$ and $\hat{F}^\ell(j, x) := \hat{f}^\ell(j)e^{2\pi i k x}$.
 (iii) Finally, $\langle F_k^{(\ell)}, \hat{F}_k^{(\ell)} \rangle = 0$ and the term $\phi(\varepsilon) : L^2(M) \rightarrow L^2(M)$ in (12) is a finite-rank operator such that $\|\phi(\varepsilon)\|/\varepsilon \xrightarrow{\varepsilon \rightarrow 0} 0$.

4.2. Proof of Theorem 4.5. To prove the above theorem, we first establish four preliminary results.

Proposition 4.6. Let $\{f^{(1)}, \dots, f^{(N)}\} \subset \mathbb{C}^N$ be the orthogonal eigenbasis from Theorem 4.2. Then, there exists $\gamma > 0$ and a family $\{f_\varepsilon^{(1)}, \dots, f_\varepsilon^{(N)}\}_{0 < \varepsilon < \gamma} \subset \mathbb{C}^N$ of unit-norm eigenbases of $P_{\varepsilon, \beta}$ such that, for every $\ell \in \{1, \dots, N\}$

- (i) the map $\varepsilon \in (0, \gamma) \mapsto f_\varepsilon^{(\ell)} \in \mathbb{C}^N$ is $\mathcal{C}^\infty$; and
- (ii) $f_\varepsilon^{(\ell)} \xrightarrow{\varepsilon \rightarrow 0} f^{(\ell)}$ in $\mathbb{C}^N$.

Proof. Fix a natural number $\ell \in \{1, \dots, N\}$. As established at the beginning of Section 4.1, by the means of perturbation theory, there exists a family $\{g_\varepsilon^{(\ell)}\}_{0 < \varepsilon < \gamma} \subset \mathbb{C}^N$ of unit-norm eigenbasis of $\mathcal{P}_{\varepsilon, \beta}$ such that (i) is satisfied.

From Theorem 4.2, it follows that $[g_\varepsilon^{(\ell)}] \rightarrow [f^{(\ell)}]$ as $\varepsilon \rightarrow 0$. Since $\|f^{(\ell)}\| = 1$ and $\|g_\varepsilon^{(\ell)}\| = 1$, we obtain that $|\langle f^{(\ell)}, g_\varepsilon^{(\ell)} \rangle| \rightarrow 1$ as $\varepsilon \rightarrow 0$. Shrinking $\gamma > 0$, if necessary, we can assume that $1/2 < \|\langle f^{(\ell)}, g_\varepsilon \rangle\|$ for $0 < \varepsilon < \gamma$.

Define

$$f_\varepsilon^{(\ell)} := \frac{\langle f^{(\ell)}, g_\varepsilon \rangle}{|\langle f^{(\ell)}, g_\varepsilon \rangle|} g_\varepsilon \text{ for every } 0 < \varepsilon < \gamma.$$

From a direct competition, we obtain that

$$\lim_{\varepsilon \rightarrow 0} \langle f_\varepsilon^{(\ell)}, f^{(\ell)} \rangle = \lim_{\varepsilon \rightarrow 0} \left\langle \frac{\langle f^{(\ell)}, g_\varepsilon \rangle}{|\langle f^{(\ell)}, g_\varepsilon \rangle|} g_\varepsilon, f^{(\ell)} \right\rangle = \lim_{\varepsilon \rightarrow 0} \frac{\langle f^{(\ell)}, g_\varepsilon \rangle \langle f_\varepsilon, g_\varepsilon \rangle}{|\langle f^{(\ell)}, g_\varepsilon \rangle|} = \lim_{\varepsilon \rightarrow 0} |\langle g_\varepsilon, f^{(\ell)} \rangle| = 1.$$

This implies that $f_\varepsilon^{(\ell)} \rightarrow f^{(\ell)}$ as $\varepsilon \rightarrow 0$, since $f_\varepsilon^{(\ell)}$ is a unit-norm vector for every $0 < \varepsilon < \gamma$. $\square$

In what follows, we establish some notation. We fix the orthonormal basis $\{f^{(1)}, \dots, f^{(N)}\} \subset \mathbb{C}^N$ given in Theorem 4.2 and a family $\{f_\varepsilon^{(1)}, \dots, f_\varepsilon^{(N)}\}_{0 < \varepsilon < \gamma} \subset \mathbb{C}^N$ of unit-norm eigenbases of $P_{\varepsilon, \beta}$ provided by Proposition 4.6. In particular, we obtain that $f_\varepsilon \xrightarrow{\varepsilon \rightarrow 0} f^{(\ell)}$ for every $\ell \in \{1, \dots, N\}$.

For each $\ell \in \{1, \dots, N\}$ and sufficiently small ε , we define $\mu_\varepsilon^{(\ell)}$ as the eigenvectors of the Hermitian conjugate $(P_{\varepsilon, \beta})^*$, which satisfy $(P_{\varepsilon, \beta})^* \mu_\varepsilon^{(\ell)} = \lambda_\varepsilon^{(\ell)} \mu_\varepsilon^{(\ell)}$. Furthermore, we impose the normalization condition

$$\langle f_\mu^{(\ell)}, \mu_\varepsilon^{(\ell)} \rangle = 1 \text{ for all } \ell \in \{1, \dots, N\} \text{ and } \varepsilon > 0. \quad (26)$$

By Theorem 3.3 for $\varepsilon > 0$ small enough, $P_{\varepsilon, \beta}$ admits N distinct (and therefore simple) eigenvalues, and by appealing to standard matrix perturbation theory, the derivatives $\dot{f}_\varepsilon^{(\ell)} = \frac{\partial}{\partial \varepsilon} f_\varepsilon^{(\ell)}$ exist for $\varepsilon > 0$. The strategy of the proof of Theorem 4.5 is to understand the behaviour of $\dot{f}_\varepsilon^{(\ell)}$ as $\varepsilon \rightarrow 0$. To achieve this, for every $\varepsilon > 0$ small enough we write $\dot{f}_\varepsilon^{(\ell)}$ in the basis

$$\{f_\varepsilon^{(1)}, \dots, f_\varepsilon^{(\ell-1)}, \mu_\varepsilon^{(\ell)}, f_\varepsilon^{(\ell+1)}, \dots, f_\varepsilon^{(N)}\},$$

i.e.

$$\dot{f}_\varepsilon^{(\ell)} = \alpha_\varepsilon^\ell \mu_\varepsilon^{(\ell)} + \sum_{m=1, m \neq \ell}^N \alpha_\varepsilon^m f_\varepsilon^{(m)}, \quad (27)$$

and we study the behaviour of the complex numbers $\alpha_\varepsilon^{(1)}, \dots, \alpha_\varepsilon^{(N)}$ as $\varepsilon \rightarrow 0$.

Observe that by repeating the proof of Theorem 4.2, but considering the matrix $(P_{\varepsilon, \beta})^* = W_\varepsilon D_{-\beta}$ instead of $P_{\varepsilon, \beta}$, and noting that the matrices $\hat{W}_L D_{-\beta}$ and $D_\beta \hat{W}_L$ have the same eigenvectors, we obtain that

$$\mu_\varepsilon^{(\ell)} \xrightarrow{\varepsilon \rightarrow 0} f^{(\ell)}. \quad (28)$$

Moreover, it is worth noting that

$$\langle f_\varepsilon^{(m)}, \mu_\varepsilon^{(\ell)} \rangle = 0 \quad \text{if } m \neq \ell. \quad (29)$$

Indeed,

$$\langle f_\varepsilon^{(m)}, \mu_\varepsilon^{(\ell)} \rangle = \frac{1}{\lambda_\varepsilon^{(m)}} \langle P_{\varepsilon, \beta} f_\varepsilon^{(m)}, \mu_\varepsilon^{(\ell)} \rangle = \frac{1}{\lambda_\varepsilon^{(m)}} \langle f_\varepsilon^{(m)}, (P_{\varepsilon, \beta})^* \mu_\varepsilon^{(\ell)} \rangle = \frac{\lambda_\varepsilon^{(\ell)}}{\lambda_\varepsilon^{(m)}} \langle f_\varepsilon^{(m)}, \mu_\varepsilon^{(\ell)} \rangle.$$

Since $\lambda_\varepsilon^{(\ell)} \neq \lambda_\varepsilon^{(m)}$, it follows that $\langle f_\varepsilon^{(m)}, \mu_\varepsilon^{(\ell)} \rangle = 0$ for $\ell \neq m$.

With the above notation and initial observations established, we proceed to prove four preliminary results, which will culminate in Theorem 4.5.

Proposition 4.7. *Let $\ell \in \{1, \dots, N\}$ and $s_\ell \in \{1, \dots, S\}$ such that $\ell \in B_{s_\ell}$, then*

$$\lim_{\varepsilon \rightarrow 0} \frac{(f_\varepsilon^{(\ell)})_j - (f^{(\ell)})_j}{\varepsilon} = \lim_{\varepsilon \rightarrow 0} \frac{(f_\varepsilon^{(\ell)})_j}{\varepsilon} = \frac{1}{e^{-2\pi i k \beta_{s_\ell}} - e^{-2\pi i k \beta_{s_j}}} \left(D_\beta \dot{W} f^{(\ell)} \right)_j$$

for each $j \in \{1, \dots, N\} \setminus B_{s_\ell}$, where $s_j \in \{1, \dots, S\}$ satisfies $j \in B_{s_j}$.

Proof. Consider $j \in \{1, \dots, N\} \setminus B_{s_\ell}$, and $s_j \in \{1, \dots, S\}$ such that $j \in B_{s_j}$. Recall from Lemma 3.4 that $(f^{(\ell)})_j = 0$. On the one hand, we obtain that

$$\begin{aligned} \frac{1}{\varepsilon} (P_{\varepsilon, \beta} f_\varepsilon^{(\ell)} - D_\beta f^{(\ell)})_j &= \frac{1}{\varepsilon} (D_\beta (W_\varepsilon - \text{Id}) f_\varepsilon^{(\ell)} + D_\beta (f_\varepsilon^{(\ell)} - f^{(\ell)}))_j \\ &= \left(D_\beta \left(\frac{W_\varepsilon - \text{Id}}{\varepsilon} \right) f_\varepsilon^{(\ell)} \right)_j + \frac{e^{2\pi i k \beta_{s_j}}}{\varepsilon} (f_\varepsilon^{(\ell)} - f^{(\ell)})_j \end{aligned}$$

On the other hand, since $f^{(\ell)} \in V_{s_\ell}$ (see (18)),

$$\begin{aligned} \frac{1}{\varepsilon} (P_{\varepsilon, \beta} f_\varepsilon^{(\ell)} - D_\beta f^{(\ell)})_j &= \frac{1}{\varepsilon} (\lambda_\varepsilon^{(\ell)} f_\varepsilon^{(\ell)} - e^{-2\pi i k \beta_{s_\ell}} f_\varepsilon^{(\ell)} + e^{-2\pi i k \beta_{s_\ell}} f_\varepsilon^{(\ell)} - e^{-2\pi i k \beta_{s_\ell}} f^{(\ell)})_j \\ &= \frac{1}{\varepsilon} (\lambda_\varepsilon^{(\ell)} - e^{-2\pi i k \beta_{s_\ell}}) (f_\varepsilon^{(\ell)})_j + \frac{e^{-2\pi i k \beta_{s_\ell}}}{\varepsilon} (f_\varepsilon^{(\ell)} - f^{(\ell)})_j. \end{aligned}$$

Combining the above equations we obtain that

$$(e^{-2\pi i k \beta_{s_j}} - e^{-2\pi i k \beta_{s_\ell}}) \left(\frac{(f_\varepsilon^{(\ell)})_j - (f^{(\ell)})_j}{\varepsilon} \right) = \frac{\lambda_\varepsilon^{(\ell)} - e^{-2\pi i k \beta_{s_\ell}}}{\varepsilon} (f_\varepsilon^{(\ell)})_j - \left(D_\beta \left(\frac{W_\varepsilon - \text{Id}}{\varepsilon} \right) f_\varepsilon^{(\ell)} \right)_j.$$

From Lemma 3.4 it follows that $(f_\varepsilon^{(\ell)})_j \xrightarrow{\varepsilon \rightarrow 0} 0$ for every $j \in \{1, \dots, N\} \setminus B_{s_\ell}$. Hence,

$$\begin{aligned} \lim_{\varepsilon \rightarrow 0} \frac{(f_\varepsilon^{(\ell)})_j - (f^{(\ell)})_j}{\varepsilon} &= \frac{1}{e^{-2\pi i k \beta_{s_j}} - e^{-2\pi i k \beta_{s_\ell}}} \left(\hat{\lambda}^{(\ell)} (f^{(\ell)})_j - \left(D_\beta \dot{W} f^{(\ell)} \right)_j \right) \\ &= \frac{1}{e^{-2\pi i k \beta_{s_\ell}} - e^{-2\pi i k \beta_{s_j}}} \left(D_\beta \dot{W} f^{(\ell)} \right)_j. \end{aligned}$$

□

Proposition 4.8. Under the above notation of equation (27), the vector $\alpha_\varepsilon = (\alpha_\varepsilon^1, \dots, \alpha_\varepsilon^N)$ satisfies $\alpha_\varepsilon = (A_\beta^\varepsilon)^{-1} \mathbf{v}_\beta^\varepsilon$ where A_β^ε is a $N \times N$ matrix such that

$$[A_\beta^\varepsilon]_{r,s} = \begin{cases} 1, & \text{if } s = \ell \\ \langle \mu_\varepsilon^{(\ell)}, \mu_\varepsilon^{(r)} \rangle, & \text{if } r \neq \ell \text{ and } s = \ell \\ \langle f_\varepsilon^{(s)}, f_\varepsilon^{(\ell)} \rangle, & \text{if } \ell = r \text{ and } s \neq \ell \\ 0, & \text{otherwise.} \end{cases}$$

and

$$\mathbf{v}_\beta^\varepsilon = \left(\langle \dot{f}_\varepsilon^{(\ell)}, \mu_\varepsilon^{(1)} \rangle, \dots, \langle \dot{f}_\varepsilon^{(\ell)}, \mu_\varepsilon^{(\ell-1)} \rangle, \langle \dot{f}_\varepsilon^{(\ell)}, f_\varepsilon^{(\ell)} \rangle, \langle \dot{f}_\varepsilon^{(\ell)}, \mu_\varepsilon^{(\ell+1)} \rangle, \dots, \langle \dot{f}_\varepsilon^{(\ell)}, \mu_\varepsilon^{(N)} \rangle \right).$$

Proof. Observe that for every $r \in \{1, \dots, \ell-1, \ell+1, \dots, N\}$ we obtain that

$$\langle \dot{f}_\varepsilon^{(\ell)}, \mu_\varepsilon^{(r)} \rangle = \left\langle \alpha_\varepsilon^\ell \mu_\varepsilon^{(\ell)} + \sum_{m=1, m \neq r}^N \alpha_\varepsilon^m f_\varepsilon^{(m)}, \mu_\varepsilon^{(r)} \right\rangle = \alpha_\varepsilon^r + \alpha_\varepsilon^\ell \langle \mu_\varepsilon^{(\ell)}, \mu_\varepsilon^{(r)} \rangle$$

and

$$\langle \dot{f}_\varepsilon^{(\ell)}, \mu_\varepsilon^{(\ell)} \rangle = \alpha_\varepsilon^\ell + \sum_{m=1, m \neq \ell}^N \alpha_\varepsilon^m \langle f_\varepsilon^{(m)}, f_\varepsilon^{(\ell)} \rangle.$$

By solving the above linear equation, we obtain the desired result. $\square$

Proposition 4.9. Let $\ell \in \{1, \dots, N\}$ and $s_\ell \in \{1, \dots, S\}$ such that $N_{s_\ell-1} < \ell \leq N_{s_\ell}$. Then,

$$\lim_{\varepsilon \rightarrow 0} \frac{(\mu_\varepsilon^{(\ell)})_j - (\mu_\varepsilon^{(\ell)})_j}{\varepsilon} = \lim_{\varepsilon \rightarrow 0} \frac{(\mu_\varepsilon^{(\ell)})_j}{\varepsilon} = \frac{1}{e^{2\pi i k \beta_{s_\ell}} - e^{2\pi i k \beta_{s_j}}} \left(\dot{W} D_{-\beta} f^{(\ell)} \right)_j$$

for each $j \in \{1, \dots, N\} \setminus B_{s_\ell}$, and $s_j \in \{1, \dots, N\}$ is such that $j \in B_{s_j}$.

Proof. The proof follows by repeating the computations in presented in Proposition 4.7 and recalling from (28) that $\mu_\varepsilon^{(\ell)} \xrightarrow{\varepsilon \rightarrow 0} f^{(\ell)}$ for every $\ell \in \{1, \dots, N\}$. $\square$

Proposition 4.10. Let $\ell \in \{1, \dots, N\}$ and $s_\ell \in \{1, \dots, S\}$ such that $\ell \in B_{s_\ell}$. Then, given $r \in \{1, \dots, N\}$ and $s_r \in \{1, \dots, S\}$ such that $r \in B_{s_r}$,

(a) if $s_r \neq s_\ell$,

$$\lim_{\varepsilon \rightarrow 0} \langle \dot{f}_\varepsilon^{(\ell)}, \mu_\varepsilon^{(r)} \rangle = \frac{1}{e^{-2\pi i k \beta_{s_r}} - e^{-2\pi i k \beta_{s_\ell}}} \left\langle D_\beta \dot{W} f^{(\ell)}, f_\varepsilon^{(r)} \right\rangle;$$

(b) if $s_r = s_\ell$ and $r \neq \ell$

$$\lim_{\varepsilon \rightarrow 0} \langle \dot{f}_\varepsilon^{(\ell)}, \mu_\varepsilon^{(r)} \rangle = \frac{1}{\hat{\lambda}^{(\ell)} - \hat{\lambda}^{(r)}} \sum_{s=1, s \neq s_\ell}^S \frac{1}{e^{-2\pi i k \beta_s} - e^{-2\pi i k \beta_{s_\ell}}} \left\langle \pi_s D_\beta \dot{W} f^{(\ell)}, \pi_s \dot{W} D_{-\beta} f^{(r)} \right\rangle;$$

where $\pi_s : \mathbb{C}^N \rightarrow V_s$ is the $\mathbb{C}^N$ -orthogonal projection onto V_s .

Proof. We start by showing (a). Differentiating the equation $P_{\varepsilon, \beta} f_\varepsilon^{(\ell)} = \lambda_\varepsilon^{(\ell)} f_\varepsilon^{(\ell)}$ with respect to ε , and applying $\langle \cdot, \mu_\varepsilon^{(r)} \rangle$ in both sides we obtain from (29) that

$$\langle \dot{P}_{\varepsilon, \beta} f_\varepsilon^{(\ell)}, \mu_\varepsilon^{(r)} \rangle + \langle P_{\varepsilon, \beta} \dot{f}_\varepsilon^{(\ell)}, \mu_\varepsilon^{(r)} \rangle = \dot{\lambda}_\varepsilon^{(\ell)} \langle f_\varepsilon^{(\ell)}, \mu_\varepsilon^{(r)} \rangle + \lambda_\varepsilon^{(\ell)} \langle \dot{f}_\varepsilon^{(\ell)}, \mu_\varepsilon^{(r)} \rangle.$$

Hence $\langle \dot{P}_{\varepsilon, \beta} f_\varepsilon^{(\ell)}, \mu_\varepsilon^{(r)} \rangle + \lambda_\varepsilon^{(r)} \langle \dot{f}_\varepsilon^{(\ell)}, \mu_\varepsilon^{(r)} \rangle = \lambda_\varepsilon^{(\ell)} \langle \dot{f}_\varepsilon^{(\ell)}, \mu_\varepsilon^{(r)} \rangle$, which implies

$$\langle \dot{f}_\varepsilon^{(\ell)}, \mu_\varepsilon^{(r)} \rangle = \frac{1}{\lambda_\varepsilon^{(\ell)} - \lambda_\varepsilon^{(r)}} \langle \dot{P}_{\varepsilon, \beta} f_\varepsilon^{(\ell)}, \mu_\varepsilon^{(r)} \rangle. \quad (30)$$

Observe that $P_{\varepsilon, \beta} = D_\beta W_\varepsilon = D_\beta (\text{Id} + \varepsilon \dot{W})$, in this way

$$\dot{P}_{\varepsilon, \beta} f_\varepsilon^{(\ell)} = D_\beta \dot{W} f_\varepsilon^{(\ell)} = \frac{1}{\varepsilon} (\lambda_\varepsilon^{(\ell)} f_\varepsilon^{(\ell)} - D_\beta f_\varepsilon^{(\ell)}). \quad (31)$$

Combining equations (30) and (31)

$$\begin{aligned} \langle \dot{f}_\varepsilon^{(\ell)}, \mu_\varepsilon^{(r)} \rangle &= \frac{1}{\lambda_\varepsilon^{(\ell)} - \lambda_\varepsilon^{(r)}} \left\langle \frac{1}{\varepsilon} (\lambda_\varepsilon^{(\ell)} f_\varepsilon^{(\ell)} - D_\beta f_\varepsilon^{(\ell)}), \mu_\varepsilon^{(r)} \right\rangle \\ &= \frac{1}{\varepsilon (\lambda_\varepsilon^{(\ell)} - \lambda_\varepsilon^{(r)})} \langle D_\beta f_\varepsilon^{(\ell)}, \mu_\varepsilon^{(r)} \rangle \\ &= \frac{1}{\varepsilon (\lambda_\varepsilon^{(\ell)} - \lambda_\varepsilon^{(r)})} \langle e^{-2\pi i k \beta_{s_\ell}} f_\varepsilon^{(\ell)}, \mu_\varepsilon^{(r)} \rangle + \frac{1}{\varepsilon (\lambda_\varepsilon^{(\ell)} - \lambda_\varepsilon^{(r)})} \langle (D_\beta - e^{-2\pi i k \beta_{s_\ell}})(1 - \pi_{s_\ell}) f_\varepsilon^{(\ell)}, \mu_\varepsilon^{(r)} \rangle \\ &= \frac{1}{\lambda_\varepsilon^{(\ell)} - \lambda_\varepsilon^{(r)}} \left\langle \frac{1}{\varepsilon} (D_\beta - e^{-2\pi i k \beta_{s_\ell}})(1 - \pi_{s_\ell}) f_\varepsilon^{(\ell)}, \mu_\varepsilon^{(r)} \right\rangle \end{aligned} \quad (32)$$

where $\pi_{s_\ell} : \mathbb{C}^N \rightarrow V_{s_\ell}$ denotes the $\mathbb{C}^N$ -orthogonal projection onto V_{s_ℓ} . From Proposition 4.7 and (28) it follows that if $r \in \{1, \dots, N\} \setminus B_{s_\ell}$

$$\begin{aligned} \lim_{\varepsilon \rightarrow 0} \langle \dot{f}_\varepsilon^{(\ell)}, \mu_\varepsilon^{(r)} \rangle &= \lim_{\varepsilon \rightarrow 0} \frac{1}{\lambda_\varepsilon^{(s_\ell)} - \lambda_\varepsilon^{(s_j)}} \left\langle (D_\beta - e^{-2\pi i k \beta_{s_\ell}}) \frac{(1 - \pi_{s_\ell}) f_\varepsilon^{(\ell)} - 0}{\varepsilon}, \mu_\varepsilon^{(r)} \right\rangle \\ &= \frac{-1}{e^{-2\pi i k \beta_{s_\ell}} - e^{-2\pi i k \beta_{s_r}}} \left\langle D_\beta \dot{W} f^{(\ell)}, f^{(r)} \right\rangle. \end{aligned}$$

implying (a).

We now show (b). From (26), (32) and Proposition 4.9 we obtain that if $s_r = s_\ell$ and $\ell \neq r$,

$$\begin{aligned} \lim_{\varepsilon \rightarrow 0} \langle \dot{f}_\varepsilon^{(\ell)}, \mu_\varepsilon^{(r)} \rangle &= \lim_{\varepsilon \rightarrow 0} \left\langle (D_\beta - e^{-2\pi i k \beta_{s_\ell}}) \frac{(1 - \pi_{s_\ell}) f_\varepsilon^{(\ell)}}{\varepsilon}, \frac{1}{(\hat{\lambda}^{(\ell)} - \hat{\lambda}^{(r)}) \varepsilon + o(\varepsilon)} \mu_\varepsilon^{(r)} \right\rangle \\ &= \frac{-1}{\hat{\lambda}^{(\ell)} - \hat{\lambda}^{(r)}} \sum_{s=1, j \neq s_\ell}^S \frac{1}{e^{-2\pi i k \beta_{s_\ell}} - e^{-2\pi i k \beta_s}} \sum_{j=N_{s-1}+1}^{N_s} \left(D_\beta \dot{W} f^{(\ell)} \right)_j \left(\overline{\dot{W} D_{-\beta} f^{(r)}} \right)_j \\ &= \frac{-1}{\hat{\lambda}^{(\ell)} - \hat{\lambda}^{(r)}} \sum_{s=1, s \neq s_\ell}^S \frac{1}{e^{-2\pi i k \beta_{s_\ell}} - e^{-2\pi i k \beta_s}} \left\langle \pi_s D_\beta \dot{W} f^{(\ell)}, \pi_s \dot{W} D_{-\beta} f^{(r)} \right\rangle. \end{aligned}$$

□

With the above results, we can finally prove Theorem 4.5

Proof of Theorem 4.5. We start by proving (i). Recall from Definition 4.1 that $\pi : v \in \mathbb{C}^N \mapsto [v] \in \mathbb{CP}^{N-1}$ denotes a $\mathcal{C}^\infty$ quotient map from $\mathbb{C}^N$ onto $\mathbb{CP}^{N-1}$. Since $[v] = [tv]$ for every $t \in \mathbb{C} \setminus \{0\}$ and $v \in \mathbb{C}^N$, from the chain rule we obtain that

$$D_v \pi(sv) = 0 \text{ for every } s \in \mathbb{C}. \quad (33)$$

Hence, from the inverse function theorem and (20) we obtain that there exists $\delta > 0$, such that for each $\varepsilon \geq 0$ small enough and every $\ell \in \{1, \dots, N\}$ the map

$$v \in \{f_\varepsilon^{(\ell)} + w : w \in \mathbb{C}^N, \langle w, f_\varepsilon^{(\ell)} \rangle = 0 \text{ and } \|w\| < \delta\} \mapsto [v] \in \mathbb{CP}^{N-1}$$

is a diffeomorphism with its image, where $f_0^{(\ell)}$ should be interpreted as $f^{(\ell)}$.

Observe that $[f_\varepsilon^{(\ell)}] = [f^{(\ell)} + \varepsilon w + \mathcal{O}(\varepsilon)]$ for some $w \in \mathbb{C}^N$ satisfying $\langle w, f^{(\ell)} \rangle = 0$ if and only if for every $\mathcal{C}^\infty$ function $g : \mathbb{CP}^{N-1} \rightarrow \mathbb{R}$ it holds that

$$g \circ [f_\varepsilon^{(\ell)}] = g \circ [f^{(\ell)}] + \varepsilon D_{[f^{(\ell)}]} g D_{f^{(\ell)}} \pi w + \mathcal{O}(\varepsilon). \quad (34)$$

In the following, we show that (34) holds when taking $w = \hat{f}^{(\ell)}$ (see (25)).

For every $\varepsilon > 0$ small enough and $\ell \in \{1, \dots, S\}$ denote $\hat{f}_\varepsilon^{(\ell)}$ as the $\mathbb{C}^N$ -orthogonal projection of $\dot{f}_\varepsilon^{(\ell)}$ (see (27) and Proposition 4.10) onto the linear space $\{v \in \mathbb{C}^N; \langle v, f_\varepsilon^{(\ell)} \rangle = 0\}$. From (26) and (29) we have that

$$\text{span} \{ \mu_\varepsilon^{(1)}, \dots, \mu_\varepsilon^{(\ell-1)}, \mu_\varepsilon^{(\ell+1)}, \dots, \mu_\varepsilon^{(N)} \} = \{ v \in \mathbb{C}^N : \langle v, f_\varepsilon^{(\ell)} \rangle = 0 \}.$$

By combining Propositions 4.8 and 4.10 with the definition of $\hat{f}^{(\ell)}$ in (25), we obtain that

$$\begin{aligned} \lim_{\varepsilon \rightarrow 0} \hat{f}_\varepsilon^{(\ell)} &= \sum_{s=1, s \neq s_\ell}^S \sum_{r \in B_{s_\ell} \setminus \{\ell\}} \frac{1}{\hat{\lambda}^{(\ell)} - \hat{\lambda}^{(r)}} \frac{1}{e^{-2\pi i k \beta_s} - e^{-2\pi i k \beta_{s_\ell}}} \left\langle \pi_s D_\beta \dot{W} f^{(\ell)}, \pi_s \dot{W} D_{-\beta} f^{(r)} \right\rangle f^{(r)} \\ &+ \sum_{s=1, s \neq s_\ell}^S \sum_{r \in B_s} \frac{1}{e^{-2\pi i k \beta_s} - e^{-2\pi i k \beta_{s_\ell}}} \left\langle D_\beta \dot{W} f^{(\ell)}, f^{(r)} \right\rangle f^{(r)} = \hat{f}^{(\ell)}. \end{aligned} \quad (35)$$

From the mean value theorem we obtain that for every $\varepsilon > 0$ small enough, there exists $c_\varepsilon \in (0, \varepsilon)$ such that

$$g \circ [f_\varepsilon^{(\ell)}] - g \circ [f^{(\ell)}] = \varepsilon \frac{d}{dt} \left(g \circ \pi \circ f_t^{(\ell)} \right) \Big|_{t=c_\varepsilon} = \varepsilon D_{[f_{c_\varepsilon}^{(\ell)}]} g D_{f_{c_\varepsilon}^{(\ell)}} \pi \dot{f}_{c_\varepsilon}^{(\ell)}.$$

By the definition of $\hat{f}_{c_\varepsilon}^{(\ell)}$, we that $\langle \hat{f}_{c_\varepsilon}^{(\ell)}, f_{c_\varepsilon}^{(\ell)} \rangle = 0$ and $\dot{f}_{c_\varepsilon}^{(\ell)} = \hat{f}_{c_\varepsilon}^{(\ell)} + b_\varepsilon f_{c_\varepsilon}^{(\ell)}$ for some $b_\varepsilon \in \mathbb{C}$. In view of equation (33) it follows that

$$D_{f_{c_\varepsilon}^{(\ell)}} \pi \dot{f}_{c_\varepsilon}^{(\ell)} = D_{f_{c_\varepsilon}^{(\ell)}} \pi (\hat{f}_{c_\varepsilon}^{(\ell)} + b_\varepsilon f_{c_\varepsilon}^{(\ell)}) = D_{f_{c_\varepsilon}^{(\ell)}} \pi \hat{f}_{c_\varepsilon}^{(\ell)}. \quad (36)$$

Hence, combining (35), (20) and (36) we obtain that

$$\lim_{\varepsilon \rightarrow 0} \frac{g \circ [f_\varepsilon^{(\ell)}] - g \circ [f^{(\ell)}]}{\varepsilon} = D_{[f^{(\ell)}]} g D_{f^{(\ell)}} \pi \hat{f}^{(\ell)}.$$

Thus, $[f_\varepsilon^{(\ell)}] = [f^{(\ell)} + \varepsilon \hat{f}^{(\ell)} + \mathcal{O}(\varepsilon)]$ and by (20), (35) and the property $\langle f_\varepsilon^{(\ell)}, \hat{f}_\varepsilon^{(\ell)} \rangle = 0$ we have $\langle f^{(\ell)}, \hat{f}^{(\ell)} \rangle = 0$, concluding the proof of (i).

In the following, we prove (ii). Since $[f_\varepsilon^{(\ell)}] = [f^{(\ell)} + \varepsilon \hat{f}^{(\ell)} + \mathcal{O}(\varepsilon)]$, $F_\varepsilon^{(\ell)} = f_\varepsilon^{(\ell)} e^{2\pi i k x}$ and $\hat{F}^{(\ell)}(j, x) := \hat{f}^{(\ell)}(j) e^{2\pi i k x}$ for every $\varepsilon > 0$ small enough, it holds that

$$\Pi_\varepsilon^{(\ell)}(\cdot) = \left\langle \cdot, F_\varepsilon^{(\ell)} \right\rangle F_\varepsilon^{(\ell)} = \frac{\left\langle \cdot, F^{(\ell)} + \varepsilon \hat{F}^{(\ell)} + \mathcal{O}(\varepsilon) \right\rangle}{\|F^{(\ell)} + \varepsilon \hat{F}^{(\ell)} + \mathcal{O}(\varepsilon)\|^2} \left(F^{(\ell)} + \varepsilon \hat{F}^{(\ell)} + \mathcal{O}(\varepsilon) \right).$$

Since $\langle F^{(\ell)}, \hat{F}^{(\ell)} \rangle = 0$, we obtain that

$$\frac{1}{\|F^{(\ell)} + \varepsilon \hat{F}^{(\ell)} + \mathcal{O}(\varepsilon)\|^2} = \frac{1}{\langle F^{(\ell)} + \varepsilon \hat{F}^{(\ell)} + \mathcal{O}(\varepsilon), \langle F^{(\ell)} + \varepsilon \hat{F}^{(\ell)} + \mathcal{O}(\varepsilon) \rangle \rangle} = 1 + \mathcal{O}(\varepsilon).$$

In this way,

$$\begin{aligned} \Pi_\varepsilon^{(\ell)}(\cdot) &= \left\langle \cdot, F^{(\ell)} + \varepsilon \hat{F}^{(\ell)} + \mathcal{O}(\varepsilon) \right\rangle \left(F^{(\ell)} + \varepsilon \hat{F}^{(\ell)} + \mathcal{O}(\varepsilon) \right) \\ &= \Pi^{(\ell)}(\cdot) + \varepsilon \left(\left\langle \cdot, F^{(\ell)} \right\rangle \hat{F}^{(\ell)} + \left\langle \cdot, \hat{F}^{(\ell)} \right\rangle F^{(\ell)} \right) + \mathcal{O}(\varepsilon). \end{aligned} \quad (37)$$

Finally, we prove (iii). Note that (i) implies $\langle F^{(\ell)}, \hat{F}^{(\ell)} \rangle = 0$. From the construction of the operator $\mathcal{O}(\varepsilon)$ in (37), it is clear that this operator is of finite rank and satisfies $\|\mathcal{O}(\varepsilon)\|/\varepsilon \xrightarrow{\varepsilon \rightarrow 0} 0$.

□

5. PROOF OF THEOREM 2.6 2(c)

From Lemma 3.2 and Theorem 4.2 we obtain that for every $\ell \in \{1, \dots, N\}$ and for every $\varepsilon > 0$ sufficiently small

$$\lambda_\varepsilon^{(\ell)} = e^{-2\pi i k \beta_s} + \varepsilon \hat{\lambda}^{(\ell)} + \mathcal{O}(\varepsilon) \text{ for some } s \in \{1, \dots, S\}. \quad (38)$$

In this section we show that there exists $\hat{\lambda}^{(\ell)} \in \mathbb{C}$ such that

$$\lambda_\varepsilon^{(\ell)} = e^{-2\pi i k \beta_s} + \varepsilon \hat{\lambda}^{(\ell)} + \varepsilon^2 \hat{\lambda}^{(\ell)} + \mathcal{O}(\varepsilon^2) \text{ for some } s \in \{1, \dots, S\}. \quad (39)$$

The proof of Theorem 2.6 is complete once the above equation is established. We first establish the following lemma before proving equation (39).

Lemma 5.1. *Let $\gamma > 0$ be as in Proposition 4.6. For every $\ell \in \{1, \dots, N\}$ the map*

$$\varepsilon \in [0, \gamma) \mapsto \begin{cases} \dot{\lambda}_\varepsilon^{(\ell)} = \frac{\partial}{\partial \varepsilon} \lambda_\varepsilon^{(\ell)}, & \text{if } \varepsilon \in (0, \gamma_k), \\ \hat{\lambda}^{(\ell)}, & \text{if } \varepsilon = 0, \end{cases} \quad (40)$$

is continuous at 0 and differentiable in $(0, \gamma)$.

Proof. Let $\ell \in \{1, \dots, N\}$ and $s \in \{1, \dots, S\}$ such that $\ell \in B_s$. Since $P_{\varepsilon, \beta} f_\varepsilon^{(\ell)} = \lambda_\varepsilon^{(\ell)} f_\varepsilon^{(\ell)}$, we obtain that

$$\dot{P}_{\varepsilon, \beta} f_\varepsilon^{(\ell)} + P_{\varepsilon, \beta} \dot{f}_\varepsilon^{(\ell)} = \dot{\lambda}_\varepsilon^{(\ell)} f_\varepsilon^{(\ell)} + \lambda_\varepsilon^{(\ell)} \dot{f}_\varepsilon^{(\ell)}.$$

By applying $\langle \cdot, \mu_\varepsilon^{(\ell)} \rangle$ in both sides, we obtain from (26) and $P_{\varepsilon, \beta}^* \mu_\varepsilon^{(\ell)} = \lambda_\varepsilon \mu_\varepsilon^{(\ell)}$ that

$$\dot{\lambda}_\varepsilon^{(\ell)} = \langle \dot{P}_{\varepsilon, \beta} f_\varepsilon^{(\ell)}, \mu_\varepsilon^{(\ell)} \rangle = \langle D_\beta \dot{W} f_\varepsilon^{(\ell)}, \mu_\varepsilon^{(\ell)} \rangle. \quad (41)$$

Since $\varepsilon \mapsto \dot{\lambda}_\varepsilon^{(\ell)}$ is differentiable in $(0, \gamma)$ (see Proposition 4.6) to conclude the proof we must prove that (40) is continuous at 0. Therefore, from (38) it is enough to show that

$$\lim_{\varepsilon \rightarrow 0} \dot{\lambda}_\varepsilon = \lim_{\varepsilon \rightarrow 0} \langle \dot{P}_{\varepsilon, \beta} f_\varepsilon^{(\ell)}, \mu_\varepsilon^{(\ell)} \rangle = \lim_{\varepsilon \rightarrow 0} \frac{\lambda_\varepsilon^{(\ell)} - e^{-2\pi i k \beta_s}}{\varepsilon} = \hat{\lambda}^{(\ell)}.$$

Observe that for every $s \in \{1, \dots, S\}$,

$$D_\beta - e^{-2i k \beta_s} = (D_\beta - e^{-2i k \beta_s})(1 - \pi_s) \quad (42)$$

where $\pi_s : \mathbb{C}^N \rightarrow V_s$ is the $\mathbb{C}^N$ -orthogonal projection onto V_s . From (31), we obtain that

$$\begin{aligned} \langle \dot{P}_{\varepsilon, \beta} f_\varepsilon^{(\ell)}, \mu_\varepsilon^{(\ell)} \rangle &= \frac{1}{\varepsilon} \langle \lambda_\varepsilon^{(\ell)} f_\varepsilon^{(\ell)} - D_\beta f_\varepsilon^{(\ell)}, \mu_\varepsilon^{(\ell)} \rangle \\ &= \frac{1}{\varepsilon} (\lambda_\varepsilon^{(\ell)} - e^{-2\pi i k \beta_s}) - \frac{1}{\varepsilon} \langle (D_\beta - e^{-2\pi i k \beta_s}) f_\varepsilon^{(\ell)}, \mu_\varepsilon^{(\ell)} \rangle, \\ &= \frac{1}{\varepsilon} (\lambda_\varepsilon^{(\ell)} - e^{-2\pi i k \beta_s}) - \frac{1}{\varepsilon} \langle (D_\beta - e^{-2\pi i k \beta_s})(1 - \pi_s) f_\varepsilon^{(\ell)}, \mu_\varepsilon^{(\ell)} \rangle. \end{aligned}$$

Since $[\mu_\varepsilon^{(\ell)}] \xrightarrow{\varepsilon \rightarrow 0} [f^{(\ell)}]$ with $f^{(\ell)} \in V_s$ and $(1 - \pi_s) f_\varepsilon^{(\ell)} / \varepsilon$ converges to an element in the orthogonal complement of V_s as $\varepsilon \rightarrow 0$ (see 4.7), we have that

$$\lim_{\varepsilon \rightarrow 0} \left\langle \frac{1}{\varepsilon} (1 - \pi_s) f_\varepsilon^{(\ell)}, \mu_\varepsilon^{(\ell)} \right\rangle = 0,$$

which implies that

$$\lim_{\varepsilon \rightarrow 0} \dot{\lambda}_\varepsilon^{(\ell)} = \lim_{\varepsilon \rightarrow 0} \langle \dot{P}_{\varepsilon, \beta} f_\varepsilon^{(\ell)}, \mu_\varepsilon^{(\ell)} \rangle = \lim_{\varepsilon \rightarrow 0} \frac{\lambda_\varepsilon^{(\ell)} - e^{-2\pi i k \beta_s}}{\varepsilon} = \hat{\lambda}^{(\ell)}.$$

□

Theorem 5.2. *Given $\ell \in \{1, \dots, N\}$, we have that*

$$\lambda_\varepsilon^{(\ell)} = e^{-2\pi i k \beta_{s_\ell}} + \varepsilon \hat{\lambda}^{(\ell)} + \varepsilon^2 \hat{\lambda}^{(\ell)} + o(\varepsilon^2),$$

where $s_\ell \in \{1, \dots, S\}$ such that $\ell \in B_{s_\ell}$, $\hat{\lambda}^{(\ell)}$ is given by Theorem 4.2 and

$$\hat{\lambda}^{(\ell)} = 2 \sum_{s=1, s \neq s_\ell}^S \frac{1}{e^{-2\pi i k \beta_{s_\ell}} - e^{-2\pi i k \beta_s}} \left\langle \pi_s D_\beta \dot{W} f^{(\ell)}, \pi_s \dot{W} D_{-\beta} f^{(\ell)} \right\rangle$$

and $\pi_s : \mathbb{C}^N \rightarrow V_s$ is the $\mathbb{C}^N$ -orthogonal projection onto V_s .

Proof. Recall from (41) that $\dot{\lambda}_\varepsilon^{(\ell)} = \langle \dot{P}_{\varepsilon, \beta} f_\varepsilon^{(\ell)}, \mu_\varepsilon^{(\ell)} \rangle = \langle D_\beta \dot{W} f_\varepsilon^{(\ell)}, \mu_\varepsilon^{(\ell)} \rangle$ for every $\varepsilon > 0$ small enough. Hence

$$\ddot{\lambda}_\varepsilon^{(\ell)} = \langle D_\beta \dot{W} \dot{f}_\varepsilon^{(\ell)}, \mu_\varepsilon^{(\ell)} \rangle + \langle D_\beta \dot{W} f_\varepsilon^{(\ell)}, \dot{\mu}_\varepsilon^{(\ell)} \rangle = \langle D_\beta \dot{W} f_\varepsilon^{(\ell)}, \dot{\mu}_\varepsilon^{(\ell)} \rangle + \langle \dot{f}_\varepsilon^{(\ell)}, (D_\beta \dot{W})^* \mu_\varepsilon^{(\ell)} \rangle.$$

From (31), (42) and

$$\langle \dot{f}_\varepsilon^{(\ell)}, \mu_\varepsilon^{(\ell)} \rangle + \langle f_\varepsilon^{(\ell)}, \dot{\mu}_\varepsilon^{(\ell)} \rangle = \frac{\partial}{\partial \varepsilon} \langle f_\varepsilon^{(\ell)}, \mu_\varepsilon^{(\ell)} \rangle = \frac{\partial}{\partial \varepsilon} 1 = 0,$$

we obtain that

$$\begin{aligned} \ddot{\lambda}_\varepsilon^{(\ell)} &= \left\langle \frac{1}{\varepsilon} (\lambda_\varepsilon^{(\ell)} f_\varepsilon^{(\ell)} - D_\beta f_\varepsilon^{(\ell)}), \dot{\mu}_\varepsilon^{(\ell)} \right\rangle + \left\langle \dot{f}_\varepsilon^{(\ell)}, \frac{1}{\varepsilon} (\overline{\lambda_\varepsilon^{(\ell)}} \mu_\varepsilon^{(\ell)} - \overline{D_\beta \mu_\varepsilon^{(\ell)}}) \right\rangle \\ &= \frac{1}{\varepsilon} \lambda_\varepsilon^{(\ell)} \left(\langle \dot{f}_\varepsilon^{(\ell)}, \mu_\varepsilon^{(\ell)} \rangle + \langle f_\varepsilon^{(\ell)}, \dot{\mu}_\varepsilon^{(\ell)} \rangle \right) - \left\langle D_\beta \dot{f}_\varepsilon^{(\ell)}, \frac{1}{\varepsilon} \mu_\varepsilon^{(\ell)} \right\rangle - \left\langle \frac{1}{\varepsilon} D_\beta f_\varepsilon^{(\ell)}, \dot{\mu}_\varepsilon^{(\ell)} \right\rangle \\ &= \left\langle -e^{-2\pi i k \beta_{s_\ell}} \dot{f}_\varepsilon^{(\ell)} + e^{-2\pi i k \beta_{s_\ell}} \dot{f}_\varepsilon^{(\ell)} - D_\beta \dot{f}_\varepsilon^{(\ell)}, \frac{1}{\varepsilon} \mu_\varepsilon^{(\ell)} \right\rangle \\ &\quad + \left\langle \frac{1}{\varepsilon} (-e^{-2\pi i k \beta_{s_\ell}} f_\varepsilon^{(\ell)} + e^{-2\pi i k \beta_{s_\ell}} f_\varepsilon^{(\ell)} - D_\beta f_\varepsilon^{(\ell)}), \dot{\mu}_\varepsilon^{(\ell)} \right\rangle \\ &= \frac{-e^{-2\pi i k \beta_{s_\ell}}}{\varepsilon} \left(\langle \dot{f}_\varepsilon^{(\ell)}, \mu_\varepsilon^{(\ell)} \rangle + \langle f_\varepsilon^{(\ell)}, \dot{\mu}_\varepsilon^{(\ell)} \rangle \right) + \left\langle (e^{-2\pi i k \beta_{s_\ell}} - D_\beta)(1 - \pi_{s_\ell}) \dot{f}_\varepsilon^{(\ell)}, \frac{1}{\varepsilon} \mu_\varepsilon^{(\ell)} \right\rangle \\ &\quad + \left\langle \frac{1}{\varepsilon} (e^{-2\pi i k \beta_{s_\ell}} - D_\beta)(1 - \pi_{s_\ell}) f_\varepsilon^{(\ell)}, \dot{\mu}_\varepsilon^{(\ell)} \right\rangle \\ &= \left\langle (e^{-2\pi i k \beta_{s_\ell}} - D_\beta)(1 - \pi_{s_\ell}) \dot{f}_\varepsilon^{(\ell)}, \frac{1}{\varepsilon} (1 - \pi_{s_\ell}) \mu_\varepsilon^{(\ell)} \right\rangle \\ &\quad + \left\langle \frac{1}{\varepsilon} (e^{-2\pi i k \beta_{s_\ell}} - D_\beta)(1 - \pi_{s_\ell}) f_\varepsilon^{(\ell)}, (1 - \pi_{s_\ell}) \dot{\mu}_\varepsilon^{(\ell)} \right\rangle. \end{aligned}$$

where the last equality follows from the fact that $(1 - \pi_s)$ is an orthogonal projection. From Propositions 4.7 and 4.9 we obtain that

$$\begin{aligned} \lim_{\varepsilon \rightarrow 0} \ddot{\lambda}_\varepsilon^{(\ell)} &= \left\langle (e^{-2\pi i k \beta_{s_\ell}} - D_\beta) \lim_{\varepsilon \rightarrow 0} (1 - \pi_{s_\ell}) \dot{f}_\varepsilon^{(\ell)}, \lim_{\varepsilon \rightarrow 0} \frac{1}{\varepsilon} (1 - \pi_{s_\ell}) \mu_\varepsilon^{(\ell)} \right\rangle \\ &\quad + \left\langle (e^{-2\pi i k \beta_{s_\ell}} - D_\beta) \lim_{\varepsilon \rightarrow 0} \frac{1}{\varepsilon} (1 - \pi_{s_\ell}) f_\varepsilon^{(\ell)}, \lim_{\varepsilon \rightarrow 0} (1 - \pi_{s_\ell}) \dot{\mu}_\varepsilon^{(\ell)} \right\rangle. \end{aligned}$$

$$\begin{aligned}
&= \sum_{s=1, s \neq s_\ell}^S (e^{-2\pi i k \beta_{s_\ell}} - e^{-2\pi i k \beta_s}) \sum_{j \in B_s} \left(\lim_{\varepsilon \rightarrow 0} \dot{f}_\varepsilon \right)_j \overline{\left(\lim_{\varepsilon \rightarrow 0} \frac{1}{\varepsilon} \mu_\varepsilon^{(\ell)} \right)_j} \\
&\quad + \sum_{s=1, s \neq s_\ell}^S (e^{-2\pi i k \beta_{s_\ell}} - e^{-2\pi i k \beta_s}) \sum_{j \in B_s} \left(\lim_{\varepsilon \rightarrow 0} \frac{1}{\varepsilon} f_\varepsilon \right)_j \overline{\left(\lim_{\varepsilon \rightarrow 0} \dot{\mu}_\varepsilon^{(\ell)} \right)_j} \\
&= 2 \sum_{s=1, s \neq s_\ell}^S \frac{1}{e^{-2\pi i k \beta_{s_\ell}} - e^{-2\pi i k \beta_s}} \sum_{j \in B_s} \left(D_\beta \dot{W} f^{(\ell)} \right)_j \overline{\left(\dot{W} D_{-\beta} f^{(\ell)} \right)_j}
\end{aligned}$$

From Lemma 5.1 the map

$$\varepsilon \in [0, \gamma) \mapsto \begin{cases} \dot{\lambda}_\varepsilon^{(\ell)} = \frac{\partial}{\partial \varepsilon} \lambda_\varepsilon^{(\ell)}, & \text{if } \varepsilon \in (0, \gamma), \\ \hat{\lambda}^{(\ell)}, & \text{if } \varepsilon = 0, \end{cases}$$

is continuous. From the Mean Value Theorem we have that for every $\varepsilon > 0$, sufficiently small, there exists $c_\varepsilon \in (0, \varepsilon)$ such that

$$\dot{\lambda}_\varepsilon^{(\ell)} - \hat{\lambda}^{(\ell)} = \ddot{\lambda}_{c_\varepsilon} \varepsilon,$$

therefore

$$\lim_{\varepsilon \rightarrow 0} \frac{\dot{\lambda}_\varepsilon^{(\ell)} - \hat{\lambda}^{(\ell)}}{\varepsilon} = 2 \sum_{s=1, s \neq s_\ell}^S \frac{1}{e^{-2\pi i k \beta_{s_\ell}} - e^{-2\pi i k \beta_s}} \left\langle \pi_s D_\beta \dot{W} f^{(\ell)}, \pi_s \dot{W} D_{-\beta} f^{(\ell)} \right\rangle$$

□

Below, we finally prove Theorem 2.6.

Proof. The proof of Theorem 2.6 follows from a direct combination of Proposition 3.7 with Theorems 4.2, 4.5 and 5.2. □

6. PROOF OF PROPOSITION 2.7

The proof of Proposition 2.7 follows from standard techniques in perturbation theory (see [15]).

Proof of Proposition 2.7. From Lemma 3.1 and [15, II.2.2], we obtain that the maps in (13) are analytic functions. Equation (14) follows from differentiating the identity $D_{k,\alpha} W_\varepsilon f_{\alpha,k,\varepsilon} = \lambda_{k,\varepsilon,\alpha} f_{k,\varepsilon,\alpha}$, with respect to α and observing, we can invert the left-hand side of the identity

$$(D_{k,\alpha} W_\varepsilon - \lambda_{k,\varepsilon,\alpha}) \frac{\partial}{\partial \alpha} f_{k,\varepsilon,\alpha} = (D_{k,\alpha} W_\varepsilon - \lambda_{k,\varepsilon,\alpha})^{-1} \left(\frac{\partial}{\partial \alpha} D_{k,\alpha} W_\varepsilon - \frac{\partial}{\partial \alpha} \lambda_{k,\varepsilon,\alpha} \right) f_{k,\varepsilon,\alpha},$$

since $\langle f_{k,\varepsilon,\alpha}, \frac{\partial}{\partial \alpha^\ell} f_{k,\varepsilon,\alpha} \rangle = 0$, for each $\ell \in \{1, \dots, N\}$.

□

7. CASE STUDY

In the section we provide a detailed example to illustrate some of our theoretical results. We consider a version of our model with $N = 33$ fibres, selecting $\dot{W}$ to be the 33×33

matrix arising from a central-difference approximation of the Laplace operator on $[0, 1]$, with Neumann boundary conditions, namely:

$$\dot{W} := \begin{pmatrix} -\frac{1}{2} & \frac{1}{2} & & & & \\ \frac{1}{2} & -1 & \frac{1}{2} & & & \\ & \frac{1}{2} & -1 & \frac{1}{2} & & \\ & & \ddots & \ddots & \ddots & \\ & & & \frac{1}{2} & -1 & \frac{1}{2} \\ & & & & \frac{1}{2} & -\frac{1}{2} \end{pmatrix} \quad (43)$$

Using $W_\varepsilon = I + \varepsilon \dot{W}$ for $\varepsilon > 0$ we obtain a family of random walks on the $N = 33$ fibres:

$$W_\varepsilon := \begin{pmatrix} 1 - \frac{\varepsilon}{2} & \frac{\varepsilon}{2} & & & & \\ \frac{\varepsilon}{2} & 1 - \varepsilon & \frac{\varepsilon}{2} & & & \\ & \frac{\varepsilon}{2} & 1 - \varepsilon & \frac{\varepsilon}{2} & & \\ & & \ddots & \ddots & \ddots & \\ & & & \frac{\varepsilon}{2} & 1 - \varepsilon & \frac{\varepsilon}{2} \\ & & & & \frac{\varepsilon}{2} & 1 - \frac{\varepsilon}{2} \end{pmatrix}. \quad (44)$$

For this choice of W_ε , it is clear that items (1) and (2) of Definition 2.4 are satisfied for any band-width vector L . Item (3) of Definition 2.4 holds because submatrices of $\dot{W}$ correspond to central difference approximations of the Laplace operator on an interval with either mixed Neumann/Dirichlet boundary conditions (for the first and last submatrix) or Dirichlet boundary conditions (for “internal” submatrices) (see e.g. [25, Section 3]). Thus, W_ε is an L -admissible family for any bandwidth vector L . For purposes of illustration we choose $S = 3$ bands rotating at incommensurate speeds given by the vector $\beta_3 = (\pi/20, e/7, 1/\sqrt{2})$, and with corresponding band-width vector $L = (11, 7, 15)$. Figure 2 displays the rotational dynamics of the model along the circular fibres.

7.1. Eigendata for $\varepsilon > 0$.

7.1.1. Eigenvalues: For $\varepsilon = \delta = 0.1$, Figure 3(left) shows the eigenspectrum of $\hat{P}_{1,\beta,L}$ (i.e. for $k = 1$). For sufficiently small ε , Proposition 2.2(2) states that these eigenvalues cluster about the 3 angles in the vector $2\pi\beta$; the latter is indicated by dotted rays. The bound (6) (including a factor $\sin(2\pi\delta)/(2\pi\delta)$) is indicated with dark blue circles and is seen to be very close to optimal. Theorem 2.5 states that for sufficiently small ε we obtain 33 distinct eigenvalues for each $k \in \mathbb{Z}$, with corresponding one-dimensional eigenspaces.

7.1.2. Eigenfunctions: Proposition 2.2 states that all eigenfunctions of $\mathcal{P}$ are of the form $F_{k,\varepsilon}^{(\ell)}(j, x) = f_{k,\varepsilon}^{(\ell)}(j)e^{2\pi i k x}$, $\ell = 1, \dots, 33; k \in \mathbb{Z}$. The magnitudes of $f_{k,\varepsilon}^{(\ell)}$ for $\ell = 1, 2, 4$ (those eigenvectors of $D_{k,\beta,L}W_\varepsilon$ corresponding to the three largest-magnitude eigenvalues) are shown in Figure 3(right). Importantly, note that because of the particular form of $F_{k,\varepsilon}^{(\ell)}(j, x)$ given above, the magnitudes of the full eigenfunctions $F_{k,\varepsilon}^{(\ell)}$ on M are constant along the circular fibres and therefore follow precisely the magnitudes given by the eigenvectors $f_{k,\varepsilon}^{(\ell)}$, such as those in Figure 3(right); see Figure 4. The upper row of Figure 4 can be compared with Figure 1(left). In both cases, the magnitudes of the complex eigenfunctions are large in the vicinity of a band of phase space rotating at an approximately common speed. Moreover, the magnitude of the eigenfunction show in Figure 1(left) is relatively constant along the approximately rotational trajectories on each wing. The lower row of Figure 4 displays the evolution of the argument of the leading eigenfunctions, which passes through a single cycle because we have chosen the lowest order of $k = 1$. Similarly in Figure 1(right), we see a single cycle corresponding to a single loop

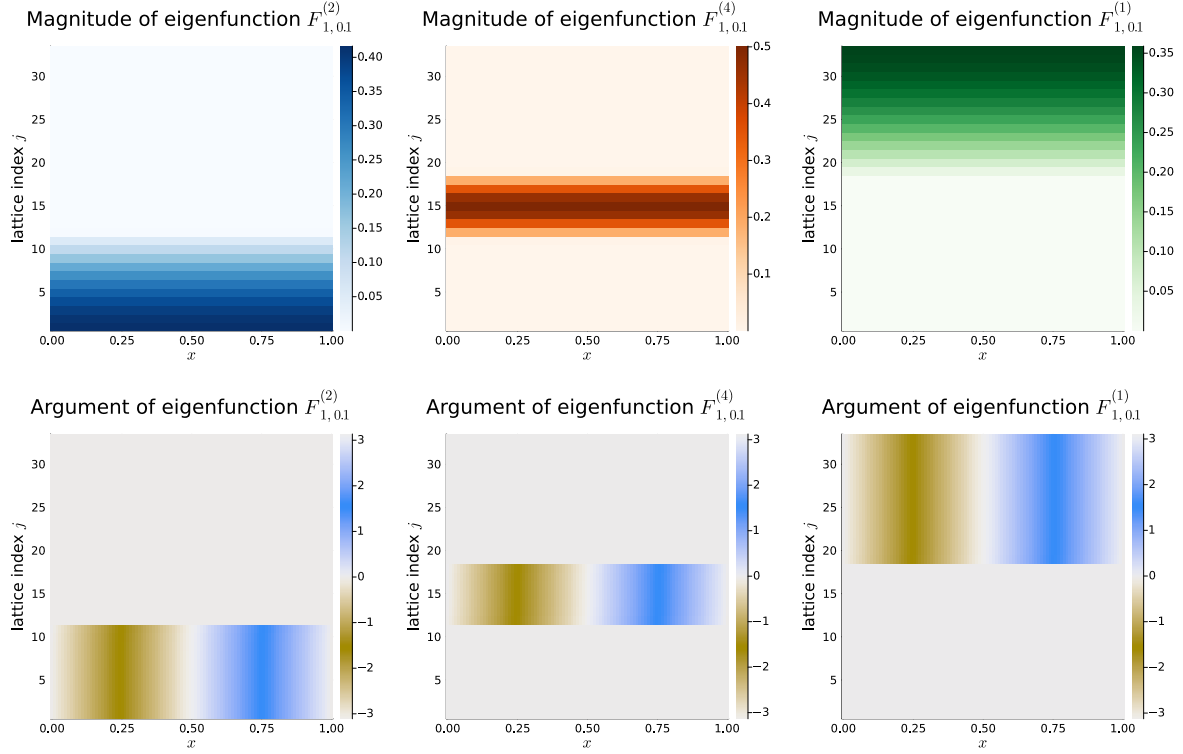


Figure 4. Details of leading eigenfunctions of $\mathcal{P}_\varepsilon$ on M , constructed from the eigenvectors in Figure 3. *Upper row:* Magnitudes of the extremal three eigenfunctions $F_{k,\varepsilon}^{(\ell)}(j, x) = f_{k,\varepsilon}^{(\ell)}(j)e^{2\pi i k x}$, $\ell = 1, 2, 4$ and $k = 1$ on $M = \{1, \dots, 33\} \times \mathbb{S}^1$ corresponding to the eigenvectors f in Figure 3, using colourschemes to match the colouring in Figure 3. Note that the support of each eigenfunction is approximately restricted to the bands associated with one of the three rotation speeds illustrated in Figure 2. *Lower row:* As for the upper row, but displaying the arguments of eigenfunctions instead of the magnitudes. Note that exactly one complex cycle occurs because $k = 1$.

around each wing of the Lorenz attractor. *Thus, the leading complex eigendata provides complete information on the various speeds or periods of rotational motion and the parts of phase space where that motion occurs.*

As $\varepsilon \rightarrow 0$, Theorem 2.5 states that we obtain a limiting orthonormal basis of eigenfunctions $\{F_k^{(\ell)}\}_{\ell=1,\dots,33;k \in \mathbb{Z}}$, and that the limiting eigenvectors $f_k^{(\ell)}$ have supports restricted to one of the three bands. This support restriction is already evident in Figure 3(right) when $\varepsilon = 0.1$.

7.2. Response of eigendata at $\varepsilon = 0$.

7.2.1. Eigenvalue response: For small ε the eigenvalues have a quadratic response by Theorem 2.6, with the linear response given explicitly in Proposition 7.1 below. Figure 5(left) shows the response of the 33 eigenvalues displayed in Figure 3(left). As stated in Theorem 2.6(2)(a) and seen explicitly in Proposition 7.1, the eigenvalues are altered in magnitude only, with the argument remaining fixed at 2π -integer multiples of the rotation speeds. Figure 5(left) demonstrates that the eigenvalues indeed move along the rays they lie on, decreasing their magnitude. Those farthest from the unit circle have the greatest response. In summary, *the argument of the eigenvalues is incredibly robust to the noise*

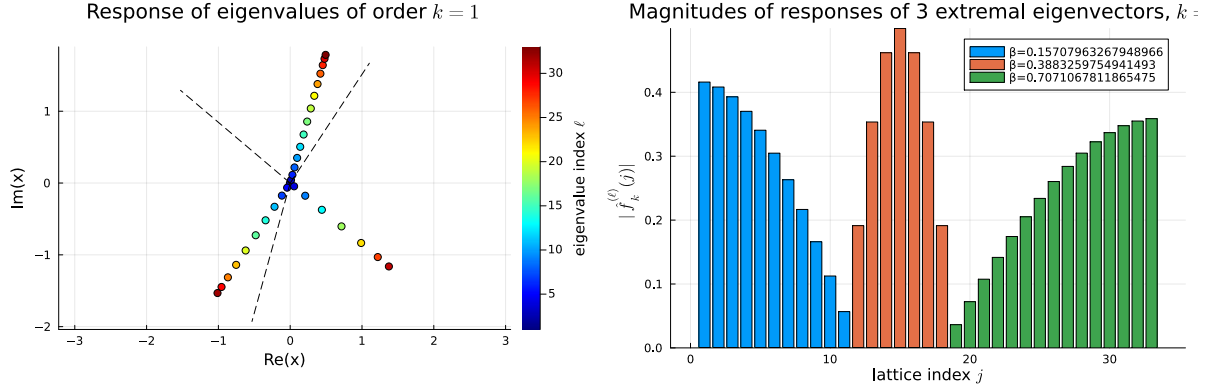


Figure 5. *Left:* The small coloured disks indicate the response of the spectrum $\hat{\lambda}_1^{(\ell)}$, $\ell = 1, \dots, 33$ of the 33×33 matrix $\hat{P}_{1,\beta,L}$, associated to the model from Figure 1, where $\beta = (\pi/20, e/7, 1/\sqrt{2})$ and $L = (11, 7, 15)$. Note that the responses of the 33 eigenvalues are grouped along three rays with L_i responses on a ray with angle $2\pi\beta_i + \pi$ for $i = 1, 2, 3$. The dashed lines and disk colouring are identical to those in Figure 3(left). *Right:* Plot of $|\hat{f}_1^{(\ell)}(j)|$ vs lattice index j for $\ell = 1, 2, 4$, where the ordering of the eigenvalues $\hat{\lambda}_1^{(\ell)}$ are shown in the left panels of this figure and Figure 3. The three response vectors have been normalised to have unit norm and correspond to the three eigenvectors displayed in Figure 3(right).

level ε , which for practical purposes means that eigenvalues of the transfer operator are robust indicators of the periods of almost-cycles.

7.2.2. Eigenfunction response: Turning now to the response of the eigenvectors, in Figure 5(right) we plot the magnitudes of responses of the three eigenvectors shown in Figure 3(right). In Figure 5(right) we can see that the responses are separately restricted to each of the three bands, and that the envelopes of the magnitudes are almost identical to those in Figure 3(right). Therefore, as in the case of the eigenvalues, these eigenvectors, and therefore the eigenfunctions of the transfer operator $\mathcal{P}_\varepsilon$, are extremely robust to the noise level ε , making them reliable indicators of the supports of distinct almost-cyclic sets in the phase space of the dynamics.

Proposition 7.1 ([25]). *Let $\hat{P}_\beta$ be the matrix defined in Theorem 4.2. The eigenvalues of $\hat{P}_\beta$ are*

$$\hat{\lambda}_{k,\beta}^{(\ell)} = \begin{cases} e^{-2\pi i k \beta_1} \left(-1 + \cos \left(\frac{2\ell - 1}{2N_1 + 1} \pi \right) \right), & \text{if } \ell \in \{1, 2, \dots, N_1\} \\ e^{-2\pi i k \beta_s} \left(-1 + \cos \left(\frac{\ell - N_{s-1}}{N_s - N_{s-1} + 1} \pi \right) \right), & \text{if } \ell \in \{N_{s-1} + 1, \dots, N_s\}, s \neq 1, S \\ e^{-2\pi i k \beta_S} \left(-1 + \cos \left(\frac{2(\ell - N_{S-1}) - 1}{2(N - N_{S-1}) + 1} \pi \right) \right), & \text{if } j \in \{N_{S-1} + 1, \dots, N\} \end{cases}$$

and their respective eigenvectors $f^{(j)}$ are:

(a) if $\ell \in \{1, \dots, N_1\}$

$$(f^{(\ell)})_j = \begin{cases} \sin \left(\pi(2\ell + 1) \left(\frac{N_1 + 1}{2N_1 + 1} - \frac{j}{2N_1 + 1} \right) \right), & \text{if } j \in \{1, \dots, N_1\} \\ 0, & \text{if } j \in \{N_1 + 1, \dots, N\} \end{cases}.$$

$$\begin{aligned}
& (b) \text{ if } \ell \in \{N_{s-1} + 1, \dots, N_s\} \text{ and } s \in \{2, \dots, S - 1\} \\
& (f^{(\ell)})_j = \begin{cases} \sin \left(\pi \frac{(j - N_{s-1})(j - N_{s-1})}{N_s - N_{s-1} + 1} \right), & \text{if } j \in \{N_{s-1} + 1, \dots, N_s\} \\ 0, & \text{if } j \in \{1, \dots, N\} \setminus \{N_{s-1} + 1, \dots, N_s\} \end{cases} \\
& (c) \text{ if } \ell \in \{N_{S-1} + 1, \dots, N\} \\
& (f^{(\ell)})_j = \begin{cases} 0, & \text{if } j \in \{1, 2, \dots, N_{S-1}\} \\ \sin \left(\pi(2\ell + 1) \frac{j - (N - N_{S-1})}{2(N - N_{S-1}) + 1} \right), & \text{if } j \in \{N_{S-1} + 1, \dots, N\} \end{cases} .
\end{aligned}$$

Observe that the quantities $\hat{\lambda}_k$ and $\hat{f}_k$ can be explicitly computed by using the above equations and Theorems 2.6.

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